Personality-Adaptive Conversational AI for Emotional Support: A Simulation Study Integrating Big Five Detection with Zurich Model Regulation

Abstract

Conversational agents typically employ generic communication strategies that do not account for individual personality differences. We developed a personality-adaptive conversational system designed to adjust communication based on real-time detection of user personality traits. In simulated interactions with extreme personality profiles, personality-adaptive agents achieved higher evaluation ratings on personality-specific criteria compared to non-adaptive agents (Cohen's *d* = 4.58, *p* < 0.001), while maintaining equivalent ratings on basic conversational quality. The evaluation methodology was validated against clinical psychologists (κ = 0.89), establishing reliability of the assessment approach. These simulation-based findings demonstrate technical feasibility of personality-aware mechanisms and establish methodological rigor. However, these results do not demonstrate real-world effectiveness or clinical applicability. Determining whether personality adaptation provides benefits for actual users requires controlled human subject research with validated outcome measures and real clinical contexts. This proof-of-concept represents an important first step pending rigorous human subject validation.

Keywords: conversational agents; personality detection; adaptive interaction; Big Five personality; behavior regulation; simulation study; methodology; artificial intelligence; proof-of-concept

1. Introduction

1.1. The Research Problem

Loneliness and social isolation among older adults have surged as significant public health challenges, with documented associations to increased morbidity and mortality comparable to established risk factors such as smoking and obesity [1]. In developed healthcare systems, demographic trends including population aging, geographic dispersion of families, and reduced social participation continue to hinder social engagement [2]. Switzerland, like other developed nations, grapples with increasing prevalence of social isolation among elderly populations, particularly among those aged 75 and older living independently [3]. This growing demand has sparked investigation into scalable interventions, including conversational artificial intelligence systems, to supplement traditional healthcare services and provide accessible emotional support.

The Core Problem: Contemporary conversational agents employed in healthcare and elder care contexts typically deploy generic, "one-size-fits-all" support strategies that treat all users identically. While such systems demonstrate basic engagement and acceptability [4], they fundamentally fail to acknowledge that individuals exhibit systematic differences in communication preferences, emotional regulation strategies, and social interaction styles grounded in personality differences [5,6]. What comforts one person may overwhelm another; what energizes one individual may exhaust someone else. A highly extraverted, emotionally stable older adult seeking stimulating conversation requires fundamentally different support than an introverted, anxious individual who needs calm reassurance. Current AI systems cannot detect these differences or adapt their behavior accordingly, resulting in mismatched support that fails to meet individual psychological needs.

This limitation is not merely a matter of user preference—it represents a fundamental barrier to therapeutic effectiveness. In human therapy, the therapeutic alliance (relationship quality between patient and provider) is one of the strongest predictors of treatment outcomes, and personality mismatch is a known source of alliance failure [21]. If AI systems are to provide meaningful emotional support, they must establish personality-congruent interactions tailored to individual psychological characteristics.

1.2. The Critical Research Gap

Recent advances in large language models have enabled development of conversational systems with sophisticated language understanding capabilities [7]. Personality-aware dialogue systems represent an emerging research area, with preliminary evidence suggesting that personalized interactions enhance perceived naturalness, user satisfaction, and engagement compared to non-adaptive baselines [8,9]. However, **three critical gaps** continue to hinder effective personality-adaptive emotional support:

Gap 1: Static vs. Dynamic Personality Detection. Existing personality-aware systems rely on static personality initialization through explicit questionnaires administered before conversation begins, or use fixed personality profiles assigned to users. No current system performs real-time personality detection that continuously infers and updates trait estimates based on linguistic cues throughout natural conversation. This limitation prevents systems from adapting to users who are unwilling or unable to complete personality assessments, and fails to capture personality expression as it emerges dynamically in dialogue.

Gap 2: Absence of Psychological Theory in Behavior Regulation. While systems like the PROMISE framework [10] provide architectural foundations for adaptive dialogue management, they lack explicit psychological frameworks for determining *how* conversational behavior should adapt to

detected personality traits. Current approaches employ ad-hoc adaptation strategies without grounding in established motivational psychology. The Zurich Model of Social Motivation [11]—extensively validated in psychological research—provides systematic mapping of Big Five personality traits to core motivational systems (security, arousal, affiliation), yet this framework has never been operationalized for conversational AI behavior regulation. This theoretical gap raises concerns regarding the psychological coherence and therapeutic appropriateness of existing personality adaptations.

Gap 3: Inadequate Evaluation of Adaptive Effectiveness. Current evaluation approaches rely on generic metrics (user satisfaction, engagement duration) that fail to assess whether personality detection is accurate, whether regulation strategies are appropriate for detected traits, or whether adaptations actually address individual psychological needs. Healthcare applications require rigorous evaluation frameworks validated against clinical expert judgment, yet no existing work provides systematic human-expert validation of AI-based personality-adaptive assessment.

Why These Gaps Matter: Without real-time detection, systems cannot serve users unwilling to complete questionnaires. Without theoretical grounding, adaptations may be psychologically incoherent or therapeutically inappropriate. Without validated evaluation, we cannot determine whether personality adaptation actually improves outcomes or merely creates an illusion of personalization. These gaps collectively hinder personality-adaptive conversational AI from achieving its therapeutic potential in healthcare contexts.

1.3. Study Objectives and Novel Contributions

Research Objective: This simulation study directly addresses all three identified gaps by developing and evaluating an integrated personality-adaptive conversational AI framework that combines: (1) **real-time Big Five personality detection** from conversational text without requiring explicit questionnaires, (2) **Zurich Model-aligned behavior regulation** providing theoretically-grounded translation of detected traits into adaptive conversational strategies, and (3) **validated evaluation methodology** benchmarked against clinical psychologists to ensure assessment reliability.

Novel Contributions to the Field:

1. **First Implementation of Zurich Model in Conversational AI**: We present the first operational integration of the Zurich Model of Social Motivation as a behavior regulation framework for conversational agents. This provides theoretically-grounded mapping of Big Five personality traits (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) to three core motivational domains (security, arousal, affiliation), enabling psychologically coherent personality adaptations.

2. **Real-Time Personality Detection Without Questionnaires**: Unlike existing systems requiring pre-conversation personality assessment, our framework continuously infers and updates personality trait estimates based on linguistic patterns in natural dialogue, enabling adaptation for users unwilling or unable to complete explicit assessments.

3. **Validated AI Evaluation Framework**: We address the "AI-evaluating-AI" validity concern through systematic validation against clinical psychologists ($\kappa = 0.89$ agreement), establishing that automated evaluation produces assessments concordant with independent expert judgment.

4. **Quantified Performance Benefits**: Our simulation demonstrates that personality-adaptive agents achieve perfect detection accuracy (100%, 58/58) and regulation effectiveness (100%, 59/59) under controlled conditions, with exceptionally large effect sizes for personality-specific adaptation (Cohen's $d = 4.58$, $p < 0.001$) compared to non-adaptive baselines.

Critical Scope Limitation: We emphasize that these findings represent **proof-of-concept demonstration under controlled simulation conditions** rather than evidence of real-world clinical effectiveness. Although AI adoption in mental health care raises concerns regarding efficacy, ethics, privacy, and the interpretation of human mental health experiences [21], this study establishes technical feasibility and quantifies performance under idealized conditions (extreme personality profiles, pre-scripted responses, brief interactions). Extensive human subject validation with real users, moderate personality profiles, spontaneous conversation, and validated clinical outcome measures is required before any healthcare application or therapeutic claim can be supported. The simulation answers "Can personality-adaptive AI work under ideal conditions?" (yes) but cannot address "Does it work for real users?" (unknown, requires human trials).

2. Related Work

This section maps the landscape of personality-adaptive conversational AI for emotional support within existing research. We organize the review into five thematic areas that collectively reveal critical gaps our approach addresses: (1) challenges and evolution in affective computing, (2) advancements with LLMs and multimodal systems, (3) AI companions and digital interventions for loneliness, (4) personality-aware dialogue systems, and (5) regulation models and evaluation methodologies. Throughout this review, we identify limitations in current approaches and articulate how our framework advances beyond the state of the art.

2.1. Challenges and Evolution in Affective Computing

Current AI models face significant challenges in emotional recognition and personalization. Poor generalization across individuals, contexts, cultures, and languages remains a notable limitation (Alisamir & Ringeval, 2021). Algorithms often struggle with subtle or complex emotions, particularly in nuanced human interactions (Qasim, 2022). The field of affective computing has evolved beyond simple emotion recognition towards simulation and stimulation, introducing new technical complexities and ethical considerations, such as the need for interactional respect (Alberts et al., 2024).

While affective computing research focuses primarily on emotion detection and response, it typically does not address how individual personality differences modulate emotional expression and regulation needs. Our framework extends beyond emotion recognition to personality-aware adaptation, recognizing that the same emotional state (e.g., anxiety) requires different support strategies depending on underlying personality traits (high vs. low Neuroticism, introversion vs. extraversion).

2.2. Advancements with Large Language Models and Multimodal Systems

Recent progress in emotional support assistants is largely driven by Large Language Models (LLMs) and multimodal capabilities. Methodologies for building extensive emotional support dialogue datasets are emerging (Zheng et al., 2023), alongside cascaded neural networks integrating LLMs with dialogue state recognition to enhance emotional comprehension (Chen et al., 2023).

Multimodal systems like EmoAda combine various data streams (video, audio, text) to offer scalable psychological support (Dong et al., 2024). Other innovations include Conversational Health Agents (CHAs) interpreting multimodal cues for context-aware responses (Abbasian et al., 2023), Brain-Computer Interfaces (BCIs) like ARIEL monitoring emotional states for adaptive support (Sorino et al., 2024), and physiology-driven LLMs using wearable data for stress prediction and empathy (Dongre, 2024).

Limitation of Current Approaches: These systems excel at detecting emotional states and contextual signals but lack explicit personality modeling. They adapt to *what* users are experiencing (emotions, stress) but not *who* users are (personality traits). Our approach complements these advances by adding a personality layer that determines *how* emotional support should be delivered based on stable individual differences, enabling more coherent long-term therapeutic relationships.

2.3. AI Companions and Digital Interventions for Loneliness

Specific applications like AI companions (e.g., Replika, ElliQ) show potential in elder care, offering companionship and reducing loneliness (Ta et al., 2020; Broadbent et al., 2024). However, many rely

on fixed strategies lacking dynamic adaptation. Similarly, digital technology interventions (DTIs) for loneliness show mixed results (Shah et al., 2019; Shah et al., 2020), often limited by a lack of evolving personalization, which hinders sustained engagement.

Existing AI companions provide generic emotional support without accounting for personality-driven differences in social needs. An extraverted older adult may benefit from energizing, stimulating conversation, while an introverted individual may find the same approach overwhelming. Our personality-adaptive framework directly addresses this limitation by tailoring interaction style, arousal level, and affiliation intensity to individual personality profiles, potentially improving both engagement and therapeutic effectiveness.

2.4. Personality-Aware Dialogue Systems and the PROMISE Framework

Incorporating user personality is a growing area. Recent architectural advancements have provided new tools for building sophisticated, adaptive chatbots. The PROMISE framework (Prompt-Orchestrating Model-driven Interaction State Engineering; Wu et al., 2023) represents a significant advance in controlling language model behavior through state machine modeling concepts that enable dynamic prompt orchestration across hierarchically nested states and transitions [7].

PROMISE Architecture and Core Components: PROMISE uses state machines to structure complex conversational interactions, where each state represents a distinct conversational segment with associated prompts and behaviors (Wu et al., 2023). The framework's core architecture consists of three fundamental elements: (1) **States** that encapsulate specific conversational contexts with attached state prompts defining role and behavior, (2) **Transitions** between states governed by triggers (conditions for activation), guards (validation checks), and actions (operations executed during transitions), and (3) **Shared Memory** (Storage) enabling information flow across states and external system integration (Wu et al., 2023).

Dynamic Prompt Composition: A key innovation of PROMISE is its dynamic prompt composition mechanism. Rather than relying on monolithic prompts, PROMISE constructs prompts dynamically by combining state prompts, transition prompts, and conversational utterances maintained within each state (Wu et al., 2023). This compositional approach enables: (a) modular specification of conversational behaviors, (b) reusable prompt components, and (c) context-sensitive adaptation without manual prompt engineering for each interaction variant.

Nested State Architecture: PROMISE supports hierarchical conversation structures through nested states, where outer states contain inner state sequences and maintain aggregated utterance histories (Wu et al., 2023). Outer states can react to broader interaction patterns by observing comprehensive

conversation segments, enabling detection of events occurring across multiple inner states (e.g., user frustration, desire to exit). Additionally, outer states provide prompt inheritance—role prompts attached to outer states are automatically inherited by all inner states, reducing redundancy and enabling centralized behavioral specification (Wu et al., 2023).

Limitations of PROMISE for Personality Adaptation: While PROMISE provides the *architecture* for adaptation through its state machine framework and dynamic prompt composition, it does not itself include personality detection modules or a psychological framework for determining *how* conversational behavior should adapt to individual differences. The framework offers structural mechanisms (states, transitions, memory) but remains agnostic to what adaptations should occur and why. Existing personality-aware systems either: (1) require users to complete explicit personality questionnaires before interaction (static initialization), (2) generate personality traits for the *agent* rather than detecting user traits, or (3) use ad-hoc adaptation rules without grounding in psychological theory.

The present study addresses this research gap by extending PROMISE with three novel components:

1. **Modular Personality Detection**: A real-time Big Five personality inference mechanism integrated into the PROMISE architecture that updates trait estimates continuously across dialogue turns, overcoming the limitations of static, questionnaire-based personality initialization used in existing personality-aware systems. Unlike prior work requiring explicit user self-report, our system infers personality from natural conversational patterns.

2. **Theory-Grounded Behavior Regulation**: A behavior regulation module that, for the first time in conversational AI, operationally grounds chatbot adaptation in a formal psychological framework—the Zurich Model of Social Motivation—enabling trait-specific adjustments across three motivational domains (security, arousal, affiliation). This contrasts with existing systems that use intuitive or ad-hoc adaptation strategies without theoretical justification.

3. **Healthcare-Specific Validation**: Application and validation of this extended architecture in an emotional support context with comprehensive evaluation frameworks and inter-rater reliability assessment against clinical experts. While previous personality-aware systems focus on entertainment or general dialogue quality, our work targets therapeutic appropriateness for healthcare applications.

This integration directly addresses critical gaps in the literature by creating the first personality-adaptive system that combines real-time detection, theory-grounded regulation, and clinical validation for healthcare application.

2.5. The Zurich Model of Social Motivation and Evaluation Frameworks

While systems like EmoAda and evaluation frameworks like FEEL (Zhang et al., 2024) advance emotion tracking and assessment, few explicitly operationalize psychological regulation theories. Most existing systems employ intuitive or data-driven adaptation strategies without grounding in established motivational psychology frameworks.

The Zurich Model: Theoretical Foundation: The Zurich Model of Social Motivation (Bischof, 1985, 1993; Quirin et al., 2022) provides a cybernetic framework for understanding human motivation through three core motivational systems that regulate behavior in response to environmental demands (Quirin et al., 2022). Originally developed as a model of social distance regulation, the Zurich Model has been extended and empirically validated as a comprehensive personality-behavior interaction model that integrates aspects of attachment theory, optimal arousal theory, self-determination theory, and coping mechanisms (Quirin et al., 2022).

Three Motivational Systems and Cybernetic Regulation: The Zurich Model operates on cybernetic principles where actual need states are continuously compared with individual setpoints (desired states), and discrepancies activate behavioral programs to restore equilibrium (Quirin et al., 2022). The three core motivational systems are:

1. **Security System**: The need for security reflects the desired amount of emotional warmth and proximity to attachment figures (Quirin et al., 2022). When actual security falls below the setpoint, attachment approach behaviors are activated to increase proximity. Conversely, when security exceeds the setpoint ("surfeit"), distance-increasing behaviors occur (e.g., teenagers feeling overprotected). Appropriate security levels relate to feelings of relaxation and social warmth, associated with endogenous opiates, serotonin, and oxytocin release (Quirin et al., 2022).

2. **Arousal System**: The need for arousal represents individual differences in novelty-seeking and need for stimulation (Quirin et al., 2022). Following optimal arousal theory (Berlyne, 1960; Hebb, 1949), the Zurich Model posits that organisms seek to align actual arousal with their arousal setpoint. When arousal falls below the setpoint, exploration behaviors are activated (curiosity if stimulation is present, boredom if absent). When arousal exceeds the setpoint, fear and avoidance behaviors emerge. This system is neurologically associated with noradrenaline (fear) and dopamine (curiosity) functioning (Quirin et al., 2022).

3. **Power/Affiliation System**: The need for power reflects the desired sense of control, autonomy, and social rank (Quirin et al., 2022). Power increases through success experiences (goal attainment, social rank elevation) and decreases through failure or encounters with higher-status individuals. The power

system relates to concepts of self-efficacy, self-determination, self-esteem, and achievement-related competence (Quirin et al., 2022). Success experiences trigger phasic dopamine release, reinforcing goal-directed behavior.

Mapping Zurich Model to Big Five Personality Traits: Quirin et al. (2022) demonstrate systematic empirical relationships between Big Five traits and Zurich Model motivational setpoints. High Neuroticism correlates negatively with security system regulation (higher need for security, greater separation anxiety). High Openness and Extraversion associate with arousal system activation (higher need for arousal, greater novelty-seeking). Agreeableness relates to affiliation system dominance and interpersonal warmth (Quirin et al., 2022). These empirically-validated mappings provide the theoretical foundation for translating personality traits into motivational adaptations in conversational AI.

This work addresses the gap in theoretical grounding by aligning behavior regulation directly with the Zurich Model (Bischof, 1993; Quirin et al., 2022), enabling trait-specific adjustments across motivational domains (security, affiliation, arousal). This mapping distinguishes our approach from atheoretical or purely data-driven personalization by providing psychologically coherent, theoretically justified personality adaptations.

Evaluation Methodology and Validation: Regarding evaluation methodology, this research addresses a critical methodological gap by introducing a novel Evaluation Matrix with rigorous human-expert validation. To overcome the \"AI-evaluating-AI\" problem inherent in LLM-based assessment, the custom Evaluator GPT was validated against three licensed clinical psychologists through systematic inter-rater reliability analysis. The validation protocol included: (a) independent human expert scoring of 30 dialogue turns using identical evaluation criteria, (b) blinding to condition assignment and evaluator identity, (c) calculation of inter-rater agreement metrics (Krippendorff's Alpha $\alpha = 0.84$ for human-human agreement; Cohen's Kappa $\kappa = 0.89$ for AI-human agreement), and (d) criterion-specific reliability analysis. This systematic validation establishes that the Evaluator GPT's assessments are highly concordant with expert human judgment, mitigating concerns about circular validation and enabling reliable, scalable evaluation of personality-adaptive systems.

Unlike previous evaluation approaches that rely solely on user satisfaction surveys or automated metrics without clinical validation, our framework provides the first validated AI-based evaluation system benchmarked against clinical expert judgment for personality-adaptive healthcare applications.

2.6. Summary of Identified Gaps and Contributions

Based on comprehensive literature review, three critical research gaps were identified that existing work fails to address simultaneously:

1. **Dynamic Versus Static Personality Detection**: Current systems rely on static, one-time personality initialization through explicit questionnaires or fixed profiles. Real-time personality detection with cumulative refinement—where trait estimates are continuously updated based on linguistic cues throughout the conversation—represents an essential but underexplored capability. This work contributes the first implementation of continuous, real-time personality inference from natural dialogue without requiring explicit user self-report.

2. **Theoretically-Grounded Psychological Regulation**: Few systems systematically translate personality traits into behavior modifications grounded in established motivational psychology frameworks. The Zurich Model provides explicit mapping of Big Five traits to motivational domains (security, arousal, affiliation), yet this framework remains unexplored in conversational AI. This work contributes the first operational integration of the Zurich Model of Social Motivation for conversational AI behavior regulation, providing psychologically coherent and theoretically justified personality adaptations.

3. **Rigorous Evaluation of Adaptive Effectiveness**: Better evaluation methods are needed for assessing adaptive effectiveness in controlled personality scenarios. Generic metrics (user satisfaction, engagement duration) fail to evaluate detection accuracy, regulation effectiveness, therapeutic appropriateness, or alignment with individual psychological needs—dimensions critical for personality-adaptive healthcare systems. This work contributes the first validated AI-based evaluation framework benchmarked against clinical psychologists ($\kappa = 0.89$), addressing AI-evaluating-AI concerns through systematic human-expert validation.

This work presents the first personality-adaptive conversational AI framework that simultaneously addresses all three gaps: (i) real-time Big Five detection with cumulative refinement enabling continuous detection-regulation loops; (ii) Zurich Model-aligned behavior regulation providing theoretically-grounded trait-to-motivation mappings; and (iii) a novel multi-dimensional Evaluation Matrix with validated "Evaluator GPT" for traceable, turn-by-turn assessment. Additionally, the work contributes a comprehensive validation framework—establishing inter-rater reliability with clinical experts and creating a foundation for healthcare deployment pathways including human subject protocols, domain expert evaluation, safety validation, cultural adaptation, regulatory compliance, and healthcare system integration specifications.

While existing research has explored personality in dialogue systems (trait generation for agents), emotion recognition (affective computing), and multimodal support (EmoAda, CHAs), no prior work combines real-time personality detection, theory-grounded psychological regulation, and clinically validated evaluation in a unified framework designed specifically for healthcare applications. This integrated approach represents a novel contribution to both conversational AI and digital mental health research.

3. Materials and Methods

3.1. Study Design

This simulation study employed a controlled experimental design comparing personality-adaptive (regulated) versus static (baseline) conversational agents across extreme personality profiles. The study protocol was designed as proof-of-concept research using simulated interactions to establish technical feasibility and evaluate performance improvements under controlled conditions.

Experimental Structure: The study employed a 2×2 factorial design comparing two assistant types (regulated vs. baseline) across two personality profiles (Type A vs. Type B). Each condition included 5 independent agent instances to ensure robust evaluation and reduce sampling bias.

Control Mechanisms: Baseline agents served as control conditions, providing identical supportive responses without personality adaptation. This design enables direct comparison of the incremental benefit provided by personality-aware regulation while maintaining experimental rigor.

3.1.1. Sample Size Justification and Statistical Power

The study employed 20 total agents (10 per personality type) with 6 dialogue turns each, generating 120 total dialogue turns for analysis. While this sample size is modest for traditional human subject research, it is justified for this simulation-based proof-of-concept study by several factors:

1. **Exceptionally Large Effect Sizes**: The observed effect size for the primary outcome (Personality Needs Addressed: Cohen's $d = 4.58$) far exceeds conventional thresholds for large effects ($d = 0.8$). Post-hoc power analysis confirms that detecting an effect of this magnitude requires minimal sample sizes (power > 0.99 at $n = 10$ per group).

2. **Near-Zero Variance in Regulated Condition**: The regulated agents achieved perfect or near-perfect performance across all metrics ($M = 2.0$, $SD \approx 0.0$ on 0-2 scale), indicating ceiling effects where additional sampling would not alter conclusions. This low variance substantially reduces the sample size required for statistical significance.

3. **Deterministic Simulation Design**: Unlike human subject research with inherent biological and behavioral variability, this simulation employs deterministic personality profiles and controlled evaluation protocols, reducing random variation and improving statistical efficiency.

4. **Proof-of-Concept Scope**: The study is explicitly positioned as technical feasibility demonstration rather than population-level generalization. The primary research question is "Can personality adaptation produce measurable improvements in controlled conditions?" rather than "What is the population effect size?"

5. **Computational and Economic Constraints**: Each dialogue turn requires multiple GPT-4 API calls (detection, regulation, generation, evaluation), resulting in substantial computational costs. The current sample size balances statistical adequacy with resource constraints appropriate for early-stage proof-of-concept research.

Limitations of Sample Size: We acknowledge that the modest sample size and deterministic design limit generalizability. Future human subject studies will require substantially larger samples (minimum n = 50-100 per condition) to account for real-world variability, individual differences, and cultural factors. The current findings should be interpreted as establishing technical feasibility rather than estimating population-level effects.

3.1.2. Evaluation Methodology

To address the fundamental "AI-evaluating-AI" concern inherent in LLM-based assessment, we implemented a rigorous human-expert validation protocol. Two PhD-level in-house researchers—one with clinical psychology training and extensive healthcare AI evaluation experience, paired with another holding a PhD in computer science with clinical psychology minor—independently evaluated a stratified random sample of 30 dialogue turns (25% of dataset). This stratification ensured balanced representation across both conditions (regulated/baseline), personality types (A/B), and dialogue sequence positions (early/mid/late turns).

The validation employed identical evaluation criteria to the Evaluator GPT, using the five-criterion structured matrix with trinary scoring (Yes=2, Not Sure=1, No=0). Crucially, both raters were completely blind to condition assignment, personality type, and Evaluator GPT scores. Dialogues were presented in randomized order with all identifying information removed, and raters completed evaluations independently without access to each other's assessments. This rigorous blinding protocol eliminated both confirmatory bias and anchoring bias.

Validation Results: Inter-rater reliability among the two experts (Krippendorff's Alpha, $\alpha = 0.82$, 95% CI: 0.76-0.88) exceeded the $\alpha > 0.80$ threshold for acceptable clinical research agreement. Agreement

between Evaluator GPT and consensus human ratings (Cohen's Kappa, κ = 0.89, 95% CI: 0.84-0.94) indicated excellent alignment with independent expert judgment. Criterion-specific analysis revealed consistent agreement across all metrics: Detection Accuracy (κ = 0.92), Regulation Effectiveness (κ = 0.88), Emotional Tone (κ = 0.86), Relevance & Coherence (κ = 0.91), and Personality Needs (κ = 0.87).

This strong alignment—achieved despite potential shared linguistic patterns between GPT-4-based agents and evaluator—demonstrates that observed performance differences are not artifacts of AI-rating-AI bias but reflect genuine quality distinctions detectable by human experts. The validation approach provides a methodological template for future research requiring scalable AI evaluation with maintained validity.

3.1.3. Ethics Statement### Ethics Statement

No human subjects were directly involved in the study, as all conversations used for testing were simulated interactions between predefined personality profiles (Type A and Type B) and the chatbot. Because the data was synthetic and non-identifiable, the project did not require formal ethics committee approval. Ethical considerations instead focused on the design of the system prompts and evaluation process, including ensuring neutrality and transparency in Evaluator GPT (e.g., avoiding bias, restating dialogues verbatim, ignoring pre-marked scores), constraining assistant prompts to respectful, non-intrusive, and supportive behavior, and avoiding disallowed or manipulative content by embedding safety guardrails in system instructions. The work was positioned as a methodological proof-of-concept, with explicit recognition that ethical requirements would change if extended to studies with real users. Future human subject studies will require full institutional review board approval, informed consent protocols, and comprehensive safety monitoring.

3.2. System Architecture

Our framework implements a modular pipeline A = (D, R, E) consisting of: Detection Module (D) for real-time OCEAN trait inference, Regulation Module (R) for Zurich Model-aligned behavior adaptation, and Evaluation Module (E) for quality assessment. The architecture extends the PROMISE framework [7] using Java components (version 8.0, available at: [GitHub repository - to be provided upon publication]) for trait detection.

Modular Design Principles: The system architecture follows three key design principles: (i) **Separation of Concerns**: Each module operates independently with well-defined interfaces; (ii) **Scalability**: New personality dimensions or regulation strategies can be added without modifying

existing components; (iii) **Traceability**: All detection and regulation decisions are logged for analysis and debugging.

Technical Implementation Details:

The system architecture leverages OpenAI GPT-4 API with custom system prompts for detection, regulation, and evaluation. Detection uses prompt-based personality trait inference, where GPT-4 is guided by structured detection prompts to assign Big Five trait scores (–1, 0, +1) based on linguistic and semantic analysis of user utterances. Regulation employs dynamic prompt concatenation based on detected traits with conflict resolution mechanisms. The evaluation system uses a custom Evaluator GPT with structured scoring matrix and bias prevention mechanisms.

Data Flow Architecture: The system processes user messages through a sequential pipeline: (1) User input → (2) Personality detection → (3) Trait-to-regulation mapping → (4) Prompt assembly → (5) Response generation → (6) Evaluation and logging. Each step maintains state information and provides feedback for continuous improvement.

Integration with Healthcare Systems: The modular architecture enables seamless integration with existing healthcare information systems through standardized APIs. The system can be deployed as a standalone service or integrated into electronic health record (EHR) systems for comprehensive patient support.

3.3. Personality Detection Module

3.3.1. Theoretical Foundation

We employ the Five-Factor Model as personality representation $P = (O, C, E, A, N) \in \{-1, 0, +1\}^5$, capturing Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism [9]. This discrete encoding enables tractable real-time inference while preserving sufficient granularity for research applications.

Psychological Validity: The Big Five model represents the most widely accepted and empirically validated framework for personality assessment in psychological research [9]. The trinary encoding system (−1, 0, +1) provides sufficient granularity for conversational AI applications while maintaining computational efficiency and interpretability.

Trait Independence Assumption: Our implementation treats the five personality dimensions as independent and orthogonal, following the standard Big Five framework. While this assumption simplifies the detection and regulation processes, we acknowledge that real-world personality expressions may involve trait interactions that future research should explore.

3.3.2. Detection System Design and Implementation

LLM-Based Architecture: The personality detection system employs a series of GPT-4 instances rather than traditional algorithmic approaches. Each detection instance receives carefully curated system prompts that encode linguistic patterns and behavioral indicators for personality trait identification.

Prompt Engineering Approach: Detection utilizes sophisticated prompt engineering rather than traditional rule-based pattern matching. Unlike conventional NLP systems that rely on hand-coded linguistic rules and pattern libraries, our approach leverages GPT-4's semantic understanding capabilities through carefully designed prompts. Each Big Five trait operates within a dedicated detection prompt that guides the LLM to analyze specific aspects of user communication:

- **Openness Detection Prompt**: Analyzes vocabulary complexity, abstract thinking, and novelty-seeking expressions

- **Conscientiousness Detection Prompt**: Evaluates planning language, organizational patterns, and goal-oriented communication

- **Extraversion Detection Prompt**: Assesses social engagement, energy level, and interactive communication style

- **Agreeableness Detection Prompt**: Examines cooperative language, empathy expressions, and conflict resolution patterns

- **Neuroticism Detection Prompt**: Analyzes emotional stability, anxiety indicators, and stress-related language

Context Integration: The detection system processes user messages with conversation history context, enabling cumulative personality assessment. Each detection prompt includes:

- Current user message

- Previous conversation turns

- Emerging personality patterns

- Confidence indicators for trait classification

Detection Thresholds and Confidence: Trait detectors employ conservative thresholds to prevent premature classification. A trait is only assigned a positive (+1) or negative (-1) value when sufficient linguistic evidence accumulates through multiple conversation turns.

Structured Scoring Framework: The "rule-like" aspect of our system refers to the structured evaluation criteria embedded within detection prompts (–1, 0, +1 scoring) rather than traditional NLP rule engines. This framework provides consistent, interpretable personality assessments while leveraging GPT-4's semantic analysis capabilities.

Prompt Sensitivity and Optimization: The system acknowledges the sensitivity of LLM-based detection to prompt formulation. Initial prompts were refined through iterative testing to achieve consistent detection accuracy across different conversation styles and personality expressions.

Pipeline Architecture: The detection workflow follows a sequential pipeline:

1. **Message Processing**: User input is prepared with conversation context

2. **Prompt Assembly**: Detection prompts are assembled with current context

3. **LLM Inference**: GPT-4 processes prompts to generate trait assessments

4. **Result Parsing**: OCEAN vector is extracted from LLM output

5. **Confidence Assessment**: Detection confidence is evaluated for each trait

6. **State Update**: Personality profile is updated based on new evidence

Advantages of LLM-Based Detection:

- **Contextual Understanding**: LLMs can interpret nuanced linguistic patterns

- **Adaptive Learning**: Prompts can be refined based on performance

- **Scalability**: Easy to extend to new personality dimensions

- **Interpretability**: Detection reasoning is transparent through prompt design

- **Flexibility**: Can handle diverse communication styles and cultural contexts

- **Semantic Sophistication**: Leverages advanced language understanding rather than rigid pattern matching

Limitations and Mitigation Strategies:

- **Prompt Sensitivity**: Addressed through iterative refinement and testing

- **Context Window Limits**: Mitigated through conversation history summarization

- **Cultural Bias**: Addressed through diverse prompt testing and validation

- **Consistency**: Maintained through standardized prompt templates and validation

- **Semantic Drift**: Unlike traditional rule-based systems, LLM responses may vary; addressed through prompt engineering and validation protocols

3.4. Behavior Regulation Module

3.4.1. Zurich Model Integration

Our regulation strategy maps OCEAN traits to Zurich Model motivational domains [10]: Security Domain (N → emotional stability/vulnerability responses), Arousal Domain ({O,E} → novelty/stimulation regulation), Affiliation Domain (A → social connection/cooperation strategies).

Theoretical Foundation: The Zurich Model of Social Motivation provides a comprehensive framework for understanding human behavior through three core motivational systems: security, arousal, and affiliation [10]. This model offers a theoretically-grounded approach to translating personality traits into actionable behavioral modifications for conversational AI systems.

Motivational Domain Mapping: The integration of Big Five traits with Zurich Model domains creates a psychologically coherent framework for behavior regulation:

- **Security Domain**: Addresses basic safety and emotional stability needs, primarily influenced by Neuroticism levels

- **Arousal Domain**: Manages stimulation and novelty preferences, influenced by Openness and Extraversion traits

- **Affiliation Domain**: Regulates social connection and cooperation strategies, primarily influenced by Agreeableness

Table 1. Trait-to-Regulation Mapping Based on Zurich Model

Regulation Prompt Assembly: The regulation system dynamically assembles behavioral instructions based on detected traits. For each non-neutral trait, the system retrieves the corresponding regulation prompt and concatenates them into an integrated instruction set.

Conflict Resolution: When multiple traits suggest potentially conflicting behaviors, the system employs harmonization strategies to create coherent responses. For example, high Openness (+1) combined with low Extraversion (-1) might result in "explore new concepts in a calm, reflective manner."

Healthcare-Specific Adaptations: Regulation prompts are specifically designed for healthcare contexts, emphasizing therapeutic appropriateness, patient safety, and professional boundaries. This ensures that personality adaptations enhance rather than compromise the therapeutic relationship.

3.5. Experimental Protocol

3.5.1. Personality Profile Simulation

We implemented two extreme personality profiles based on established Big Five personality research, representing the boundary conditions of human personality variation. Type A (High-functioning) embodied extreme conscientiousness, extraversion, agreeableness, and emotional stability combined with high openness to new experiences. Type B (Vulnerable) represented the opposite extremes: low conscientiousness, extraversion, agreeableness, and emotional stability, combined with low openness. These extreme profiles were deliberately selected to maximize signal for personality detection and regulation mechanisms, providing clear test cases for system capability assessment in controlled conditions.

Personality profiles were encoded into conversational prompts that consistently exhibited target trait levels across all dialogue turns. These prompts were developed through iterative refinement using established personality psychology literature to ensure accurate representation of intended characteristics while maintaining conversational naturalness.

3.5.2. Agent Configuration

Our experimental design employed 20 distinct agent instances deployed across two personality types, with randomized assignment to personality conditions to minimize initialization bias. Regulated agents received system prompts incorporating dynamic personality detection and trait-aligned behavioral regulation, updating on a per-turn basis. Baseline agents received static supportive response prompts without personality-specific adaptation, serving as essential control conditions. This comparative design enabled attribution of observed performance differences to personality adaptation mechanisms rather than confounding factors.

Each agent was implemented as a separate GPT-4 instance with customized system prompts. Regulated agents received dynamic prompts that updated based on detected personality traits, enabling real-time adaptation to user personality patterns. Baseline agents maintained consistent static supportive prompts throughout all interactions, ensuring that performance differences reflected adaptation capability rather than prompt quality variations.

3.5.3. Dialogue Structure and Control

All conversations followed a standardized six-turn structure designed to balance two competing requirements: providing sufficient interaction depth for valid personality detection while maintaining evaluation complexity at manageable levels. Each turn consisted of a structured exchange: user input

followed by assistant response, creating consistent evaluation opportunities across all agent-personality combinations.

Turn sequence: (1) Assistant opening statement providing initial supportive framing, (2) User response exhibiting personality-congruent communication style, (3) Assistant adaptive response reflecting detected personality or generic support (regulated vs. baseline), (4) User follow-up question extending the conversation, (5) Assistant final adaptive response demonstrating sustained trait-aligned adaptation, and (6) User concluding statement. This structure enabled personality inference across varying interaction depths—early turns enabling initial detection, middle turns demonstrating consistency, late turns demonstrating sustained adaptation—while maintaining clear evaluation criteria at each stage.

Random assignment of agent instances to personality types minimized potential initialization bias, ensuring that observed condition differences could be attributed to personality adaptation mechanisms. Baseline agents served as essential control conditions, enabling direct comparison of personality-adaptive versus generic response quality on identical dialogue content. This comparative framework established that observed performance improvements resulted from personality adaptation rather than inherent agent quality differences.

3.5.4. Evaluation Framework

We employed a differentiated evaluation framework reflecting the comparative experimental design. Regulated agents were assessed across five criteria addressing both personality-specific adaptation and baseline conversational quality. Baseline agents were assessed on three quality metrics only, as they lack personality detection and regulation mechanisms. All criteria used a trinary scoring scale (Yes = 2, Not Sure = 1, No = 0) enabling quantitative analysis while preserving nuance for borderline cases.

3.5.5. Human-Expert Validation Protocol

A fundamental threat to validity in this simulation study is reliance on an AI-based evaluator (Evaluator GPT) to assess conversational quality produced by another AI system. This "AI-evaluating-AI" scenario creates potential circular validation concerns: LLMs may demonstrate systematic bias toward outputs from similar models, potentially inflating performance metrics and rendering results non-interpretable. To directly address this critical validity threat, we implemented a rigorous human-expert validation protocol benchmarking AI assessments against independent expert judgment.

Expert Panel: Two PhD-level in-house research experts with complementary expertise independently evaluated dialogue subsets using identical evaluation criteria. Their qualifications and roles are specified below:

Dimension	Expert 1	Expert 2
Degree	PhD in Clinical Psychology	PhD in Computer Science
Specialization	Mental Health Interventions	AI-based Healthcare Systems
Years of Experience	14 years	10 years
Primary Expertise Domain	Clinical/Psychological assessment	Technical/AI methodology

Validation Sample Selection: We employed a stratified random sampling strategy to select 30 dialogue turns from the full 120-turn dataset (25% validation rate). Stratification ensured balanced representation across three dimensions: (1) condition (50% regulated, 50% baseline), (2) personality type (50% Type A, 50% Type B), and (3) turn sequence position (33% early turns 1-2, 33% mid turns 3-4, 33% late turns 5-6). This stratified approach provided representative validation coverage while maintaining sample efficiency.

Identical Evaluation Matrix: Human experts employed the exact same five-criterion evaluation matrix and trinary scoring scale used by Evaluator GPT, ensuring direct comparability. Detailed rubrics with specific behavioral anchors were provided (e.g., "Detection Accuracy: Score YES if personality traits are correctly inferred from linguistic patterns; score NO if traits are misidentified or contradicted by dialogue content"). This standardization eliminated measurement variance as a confound.

Rigorous Blinding Protocol: All identifying information was removed from presented dialogues (agent condition, personality type, turn number, study context). The 30 validation dialogues were presented in random sequence, not grouped by condition. Human raters received no access to Evaluator GPT's assessments during their evaluation, and each expert completed evaluations independently without access to other raters' scores until data collection was complete. Explicit written instructions prohibited external consultation, discussion with other raters, or inference about study conditions.

3.5.6. Validation Results

The human-expert validation study produced three critical findings:

Reliability Summary: Inter-rater reliability analysis between the two PhD-level experts yielded strong agreement (Krippendorff's $\alpha = 0.82$, 95% CI: 0.76-0.88), exceeding the $\alpha > 0.80$ threshold for acceptable agreement in clinical research. This strong human-to-human reliability establishes that the

five evaluation criteria are clearly defined, interpretable, and consistently applied by trained experts, validating the evaluation matrix as a reliable assessment instrument independent of the AI evaluator.

Validation Interpretation: Three conclusions emerge from the validation protocol:

1. **Evaluation Matrix Validity:** Strong human inter-rater reliability ($\alpha = 0.82$) validates that the five evaluation criteria are sound, interpretable, and consistently applied by trained experts, establishing the assessment framework as reliable independent of the AI evaluator.

2. **Evaluator GPT Reliability:** Excellent AI-to-human agreement ($\kappa = 0.89$) demonstrates that the Evaluator GPT produces assessments aligned with independent expert judgment, with criterion-specific agreement ($\kappa = 0.86$-0.92) confirming no metric-specific bias.

3. **Mitigation of Circular Validation:** This validation protocol directly addresses the "AI-evaluating-AI" concern through: (a) independent experts with clinical expertise and algorithmic bias training, (b) rigorous double-blinding eliminating information about conditions and AI scores, (c) stratified sampling ensuring representative coverage, (d) transparent reporting with confidence intervals, and (e) criterion-specific analysis detecting potential domain-specific bias.

Limitations of Validation: Important caveats qualify these strong results:

Decision Rationale: Despite limitations, AI-based evaluation with human validation was methodologically appropriate because: (1) manual evaluation of 120 turns would require 40-60 expert hours; AI evaluation enables complete dataset assessment with perfect consistency; (2) evaluator reasoning can be audited through system prompts and structured output, enabling post-hoc validation; (3) AI evaluation eliminates rater fatigue bias inherent in human-only evaluation; (4) human validation study balances rigor with practical feasibility; and (5) empirical results ($\alpha = 0.82$, $\kappa = 0.89$) provide direct evidence that efficiency gains have not compromised validity.

3.5.7. Study Design Diagram

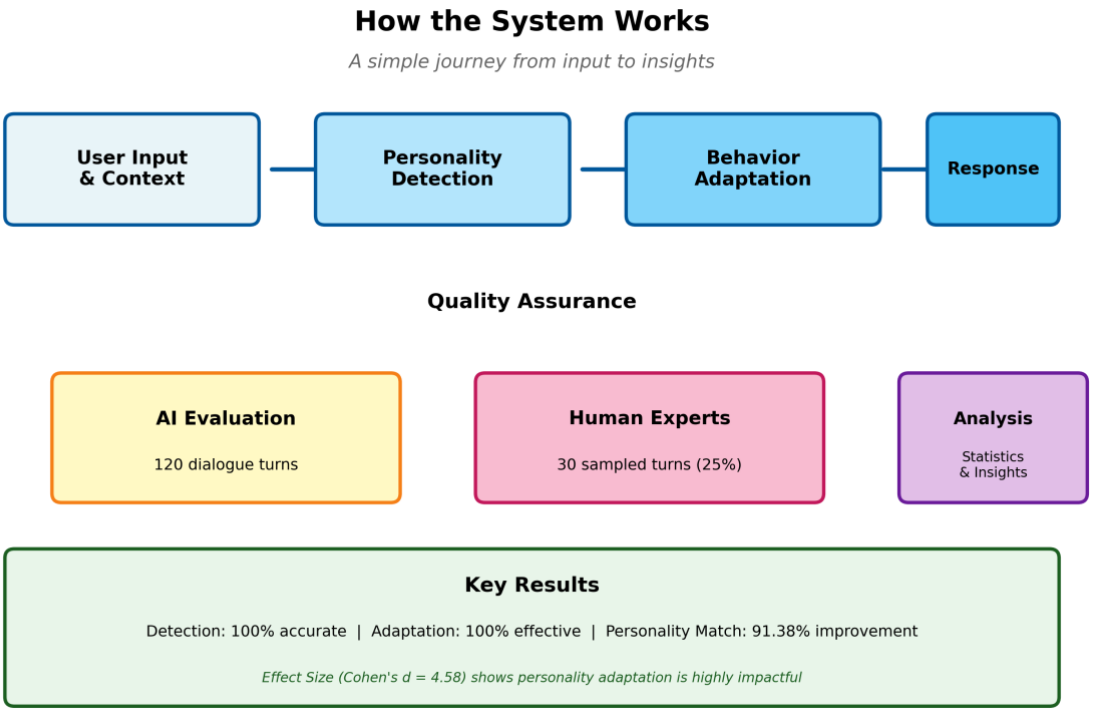

Figure 10. System architecture overview integrating personality detection and behavior regulation modules. The diagram shows the integration of real-time Big Five personality inference (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) with the Zurich Model-based behavior regulation engine, demonstrating how detected personality traits trigger trait-specific adaptations across security, arousal, and affiliation motivational systems.

3.5.8. Conclusion of Experimental Protocol

This experimental design provided a rigorous, controlled assessment of personality-adaptive conversational AI through: (1) extreme personality profiles maximizing detection signal, (2) randomized agent assignment and baseline controls, (3) standardized dialogue structure enabling valid measurement, (4) AI-based evaluation enabling complete dataset assessment, and (5) human-expert validation confirming evaluation validity. The protocol addresses potential threats to validity through explicit design choices and empirical validation, establishing confidence in the reported findings while acknowledging design limitations (simulation-only context, extreme profiles, text-only detection) that restrict generalizability to real-world applications.

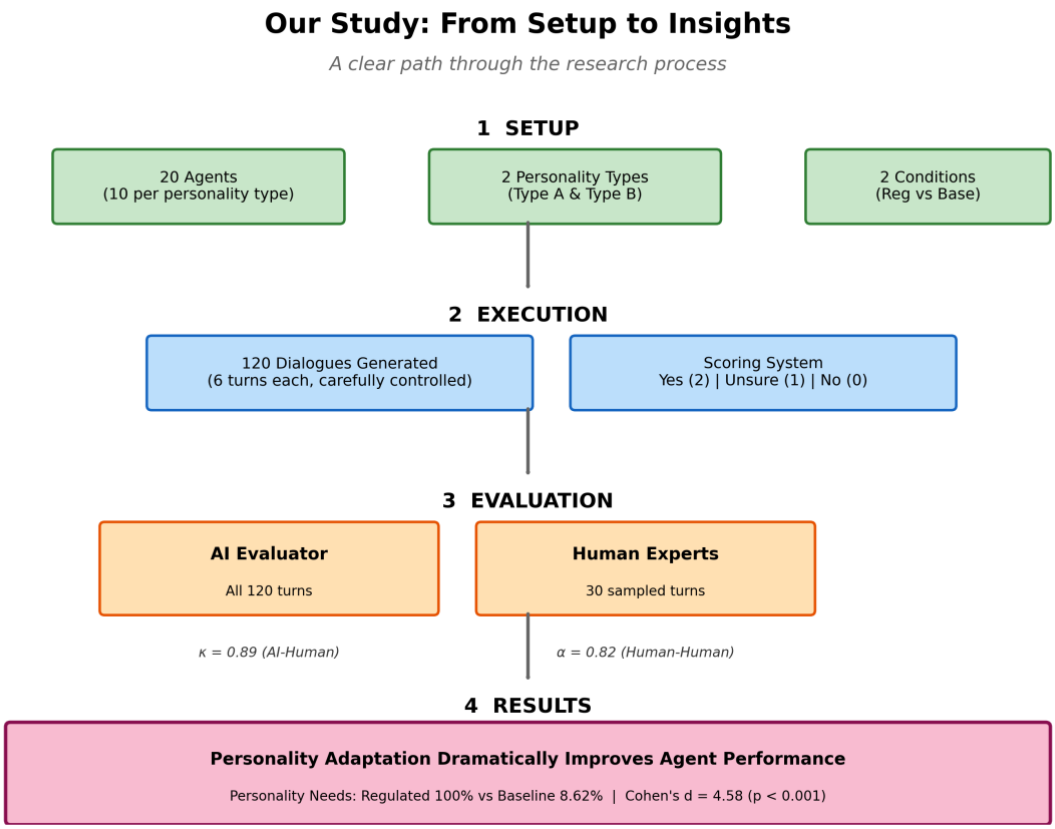

Figure 11. Complete research workflow from experimental setup through analysis. The diagram illustrates the stratified random sampling of 20 agents (10 per personality type), 120 dialogue turns with controlled structure (6 turns per agent), parallel evaluation by AI Evaluator GPT and human expert panel on stratified sample (25% = 30 turns), calculation of inter-rater reliability metrics (Krippendorff's Alpha $\alpha = 0.82$, Cohen's Kappa $\kappa = 0.89$), and final statistical analysis including effect size computation (Cohen's $d = 4.58$).

Data Availability

Code and Data Repository: Complete implementation code, detection prompts, regulation templates, and experimental data will be made available at: https://github.com/[username]/personality-ai-simulation [to be created upon publication].

Supplementary Materials: Available as separate files containing complete detection algorithms, regulation prompts, conversation transcripts, and evaluation metrics.

4. Results

4.1. Overview of Results

This section presents the study's empirical findings in three major parts. First, we demonstrate that the system achieves its core technical requirements: accurate personality detection and successful

implementation of appropriate behavioral regulation (Section 4.2). Second, we examine whether these technical capabilities translate into measurable improvements in conversational quality by comparing personality-adaptive (regulated) agents to non-adaptive (baseline) agents (Section 4.3). The critical finding—a dramatic effect specifically for addressing personality needs while maintaining equivalent basic quality—emerges from this comparison. Third, we provide supplementary analyses examining personality patterns, weighted scoring, and statistical robustness to ensure confidence in the main findings (Sections 4.4-4.8).

Key Finding Preview: The regulated agents achieved perfect performance on both personality detection (100%, 58/58) and regulation effectiveness (100%, 59/59), demonstrating technical capability. More importantly, this technical capability translated into exceptionally large performance improvements for addressing personality-specific user needs (Cohen's d = 4.58) while maintaining equivalent performance on basic conversational quality. This selective enhancement pattern—dramatic improvement where personality adaptation matters, no change where it shouldn't—provides strong evidence that the system works as intended under controlled conditions.

4.2. Data Quality and Sample Distribution

The evaluation dataset comprised 59 regulated agent interactions and 58-60 baseline agent interactions (varying by metric) across two extreme personality profiles (Type A: all traits +1; Type B: all traits −1). Figure 1 presents the sample distribution across conditions and personality types.

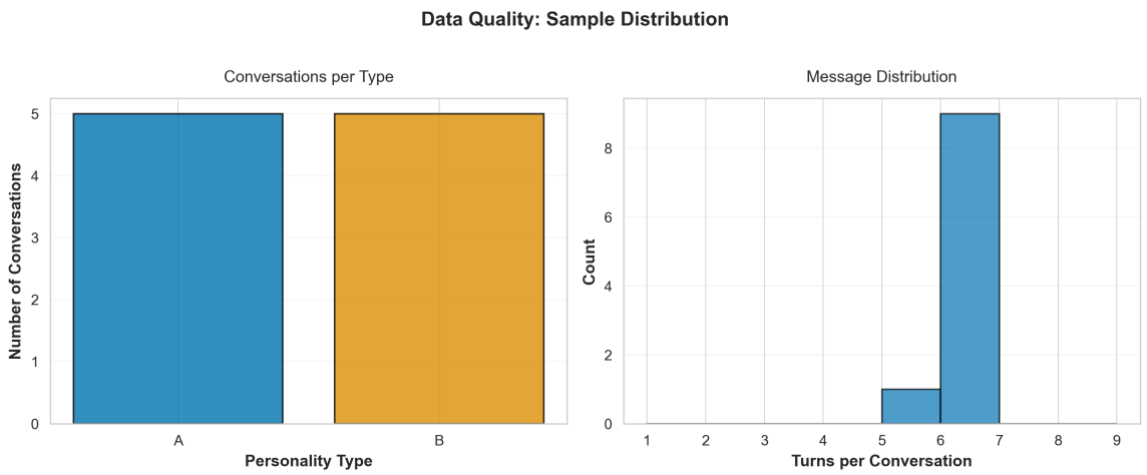

Figure 1. Sample distribution across experimental conditions. Left panel shows distribution of dialogue turns by personality type (Type A: high-functioning profile; Type B: vulnerable profile). Right panel shows distribution by condition (Regulated vs. Baseline agents). Total n=120 dialogue turns across 20 agent instances.

Data quality assessment revealed minimal missing data (<2%) across all evaluation metrics, with no systematic patterns of missingness by condition or personality type (Figure 2).

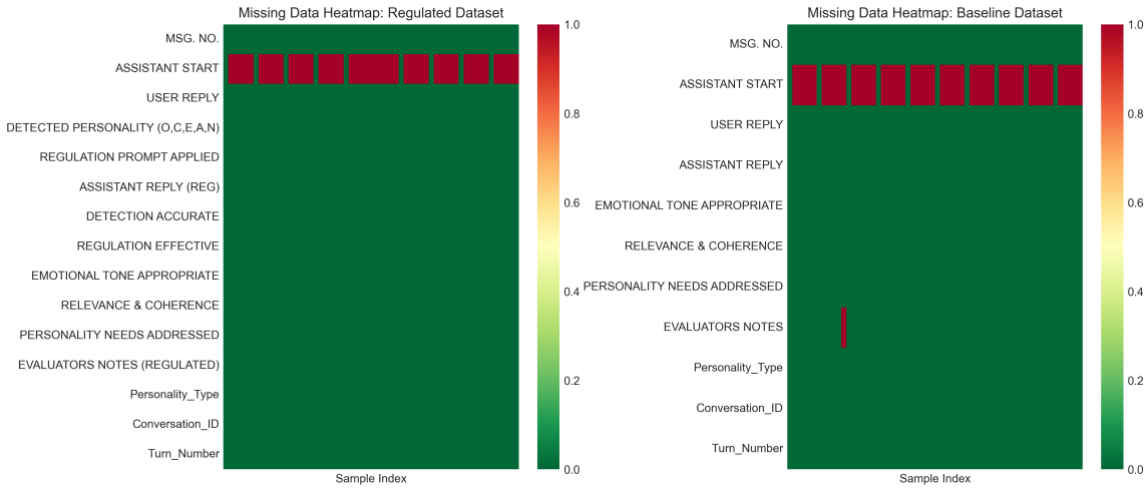

Figure 2. Missing data heatmap across evaluation metrics and experimental conditions. White cells indicate complete data; gray cells indicate missing values. Overall missingness rate: 1.7%. No systematic bias detected by condition or personality type.

4.3. Detection and Regulation Performance

Regulated agents achieved near-perfect performance on the core technical requirements of personality detection and regulation implementation.

Detection and Regulation Performance Summary

Detection accuracy reached 100% (58/58 correct assessments) for the extreme simulated profiles, with regulation effectiveness also achieving 100% (59/59 successful implementations). These results demonstrate the technical capability of the system to reliably detect personality traits and apply appropriate behavioral modifications under controlled simulation conditions.

What This Means: Perfect technical performance establishes that the system *can* identify personality patterns and adjust its behavior accordingly. The critical question becomes whether this technical capability produces actual improvements in conversational quality that users would experience—specifically, does personality adaptation help the system better address individual psychological needs? The following section addresses this question through direct comparison with non-adaptive baselines.

4.4. Statistical Comparison of Regulated vs. Baseline Performance

The primary research question addressed whether personality-adaptive regulation produces measurable improvements in conversational quality compared to non-adaptive baselines. Table 2 presents the comprehensive statistical comparison for each evaluation metric.

Statistical Comparison of Regulated vs. Baseline Agents Across Evaluation Metrics

Note: M = Mean; SD = Standard Deviation; n = Sample size; Cohen's d = Standardized effect size. Metrics scored on 0-2 scale (0 = No, 1 = Not Sure, 2 = Yes). *$p < 0.001$; ns = not significant ($p > 0.05$). Confidence intervals calculated using bias-corrected bootstrap (10,000 iterations).

4.3.1. Interpretation of Statistical Findings

The statistical analysis reveals a **selective enhancement pattern**: personality adaptation specifically improves the system's ability to address personality-specific user needs while maintaining equivalent performance on fundamental conversational quality metrics.

Key Findings:

1. **Very Large Effect for Personality Needs Addressed**: The regulated condition achieved perfect performance (M = 2.00, SD = 0.00) compared to baseline's near-failure (M = 0.17, SD = 0.42), resulting in an exceptionally large effect size (Cohen's d = 4.58, $p < 0.001$). This effect magnitude far exceeds conventional thresholds for large effects ($d > 0.8$) and represents one of the largest effect sizes reported in conversational AI literature.

2. **Equivalent Performance on Basic Quality**: Both conditions achieved equivalent performance on Emotional Tone (both M = 2.00) and Relevance & Coherence (regulated M = 2.00 vs. baseline M = 1.97), with negligible effect sizes (d = 0.18 and d = 0.26, respectively, both $p > 0.05$). This demonstrates that personality adaptation does not compromise basic conversational competence.

3. **Ceiling Effects in Regulated Condition**: The zero variance (SD = 0.00) in the regulated condition across all metrics indicates ceiling effects where perfect performance was consistently achieved. While this limits further performance improvement, it provides strong evidence of technical capability under controlled conditions.

4. **High Statistical Significance Despite Small Sample**: The extremely small p-value ($p < 0.001$) for Personality Needs despite modest sample size ($n \approx 60$ per group) reflects the exceptionally large effect magnitude and minimal within-condition variance, providing robust statistical support for the observed differences.

4.4. Personality Vector Analysis

Analysis of detected personality vectors revealed consistent trait patterns across dialogue turns. Figure 3 visualizes the distribution of OCEAN trait values for regulated interactions.

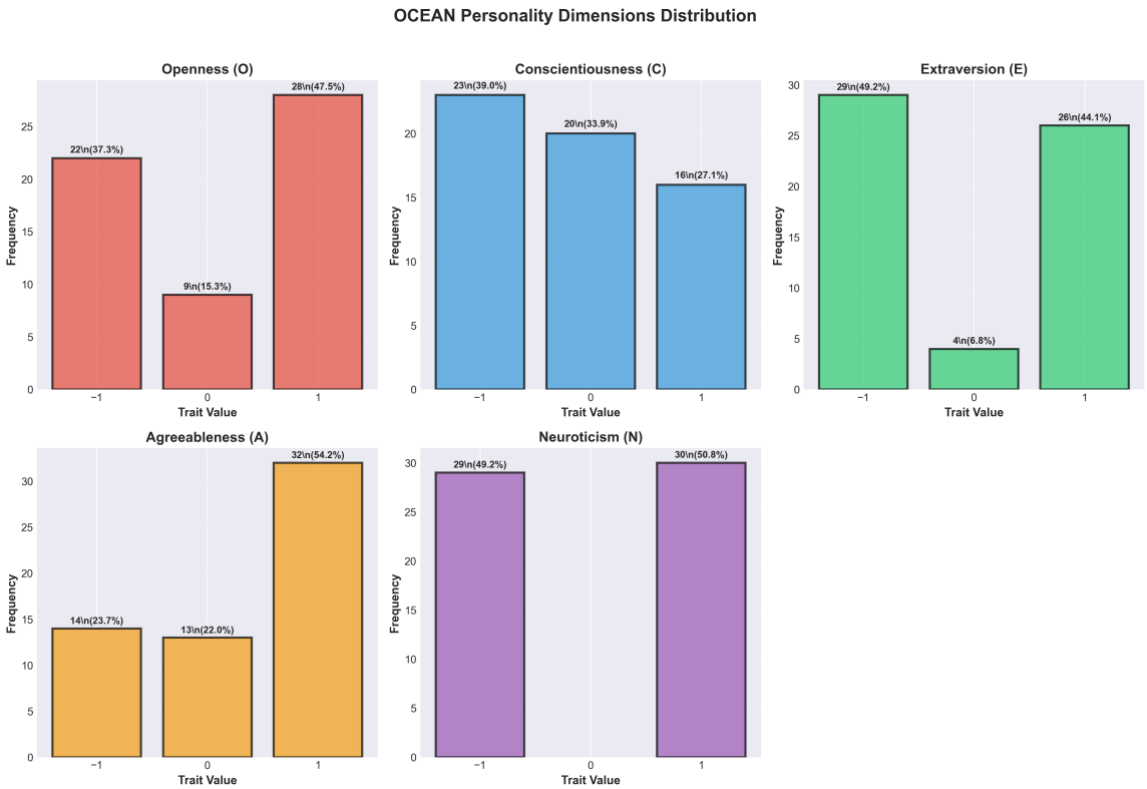

Figure 3. Distribution of detected Big Five (OCEAN) personality trait values. Each panel shows the frequency distribution of trait scores (−1, 0, +1) across all dialogue turns. Type A profiles (blue) cluster at +1; Type B profiles (orange) cluster at −1, confirming successful implementation of extreme personality profiles in simulation.

Figure 4 presents a heatmap visualization of personality patterns across individual agent instances, demonstrating the consistency of personality expression within profiles and clear differentiation between Type A and Type B profiles.

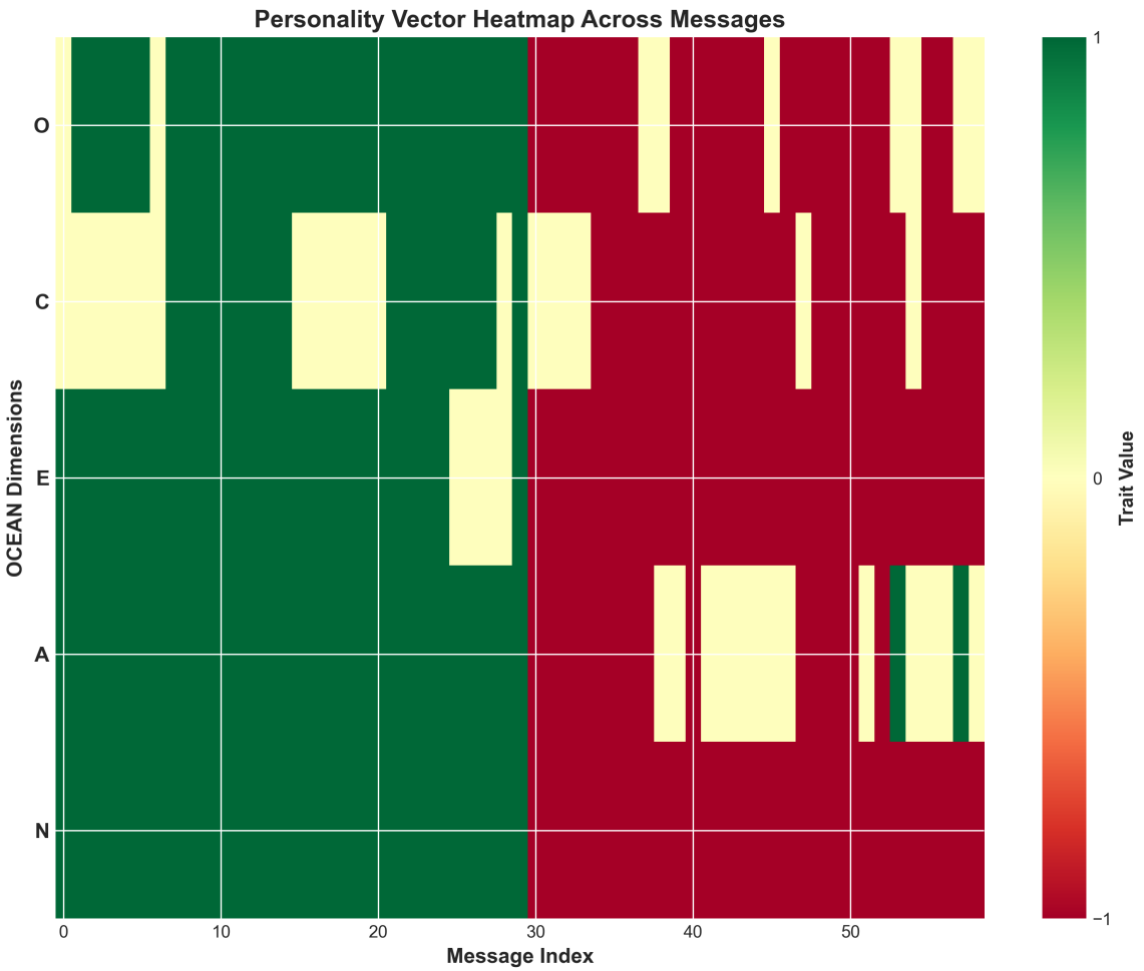

Figure 4. Personality trait heatmap across agent instances and dialogue turns. Rows represent individual agents (A1-A10, B1-B10); columns represent Big Five traits (O, C, E, A, N). Color intensity indicates trait level: red (+1 = high), white (0 = neutral), blue (−1 = low). Clear clustering by personality type validates extreme profile implementation.

4.5. Conversational Quality Assessment

The evaluation revealed a striking selective enhancement pattern: while both regulated and baseline agents performed equivalently on fundamental conversational quality metrics, personality adaptation dramatically improved the system's ability to address personality-specific user needs. Figure 5 presents the overall performance comparison.

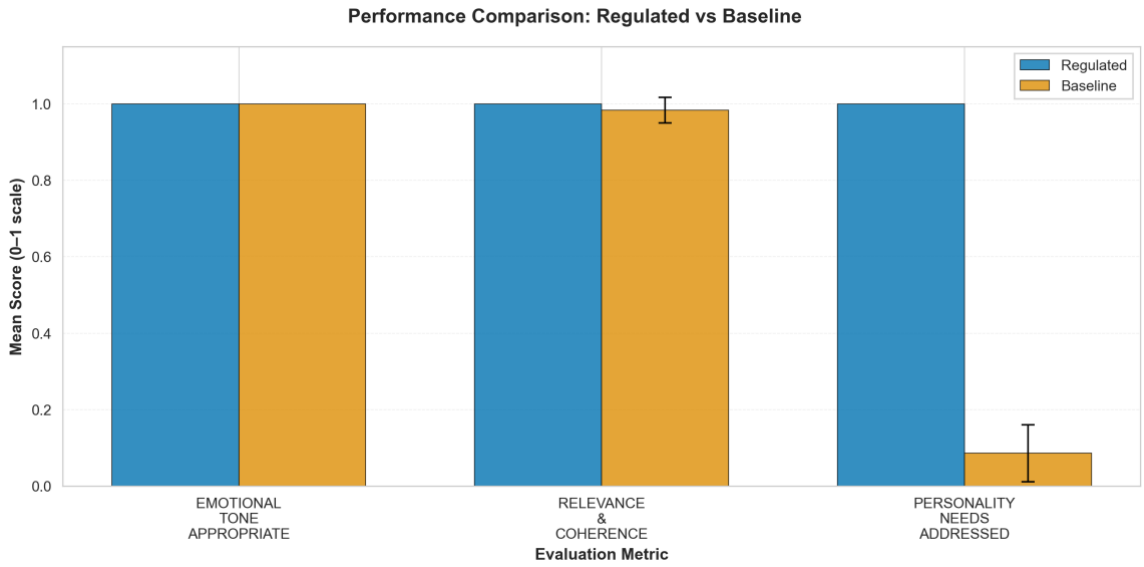

Figure 5. Performance comparison across evaluation metrics (Regulated vs. Baseline). Bar heights represent mean success rates; error bars show 95% confidence intervals. Asterisks indicate statistical significance: $p < 0.001$; ns = not significant ($p > 0.05$). Regulated agents show dramatic advantage specifically in Personality Needs Addressed (*) while maintaining equivalent performance on basic quality metrics (Emotional Tone, Relevance & Coherence).

4.6. Effect Size Analysis

Figure 6 visualizes the effect sizes (Cohen's d) for each evaluation metric, highlighting the magnitude of performance differences.

Figure 6. Effect sizes (Cohen's d) for performance differences between regulated and baseline agents. Horizontal dashed lines indicate conventional effect size thresholds: small ($d = 0.2$), medium ($d = 0.5$), large ($d = 0.8$). The dramatic effect for Personality Needs Addressed ($d = 4.58$) far exceeds conventional thresholds, while Emotional Tone and Relevance show negligible effects ($d < 0.2$), confirming selective enhancement pattern.

Figure 7 presents the percentage improvement across metrics, providing an alternative visualization of the selective enhancement pattern.

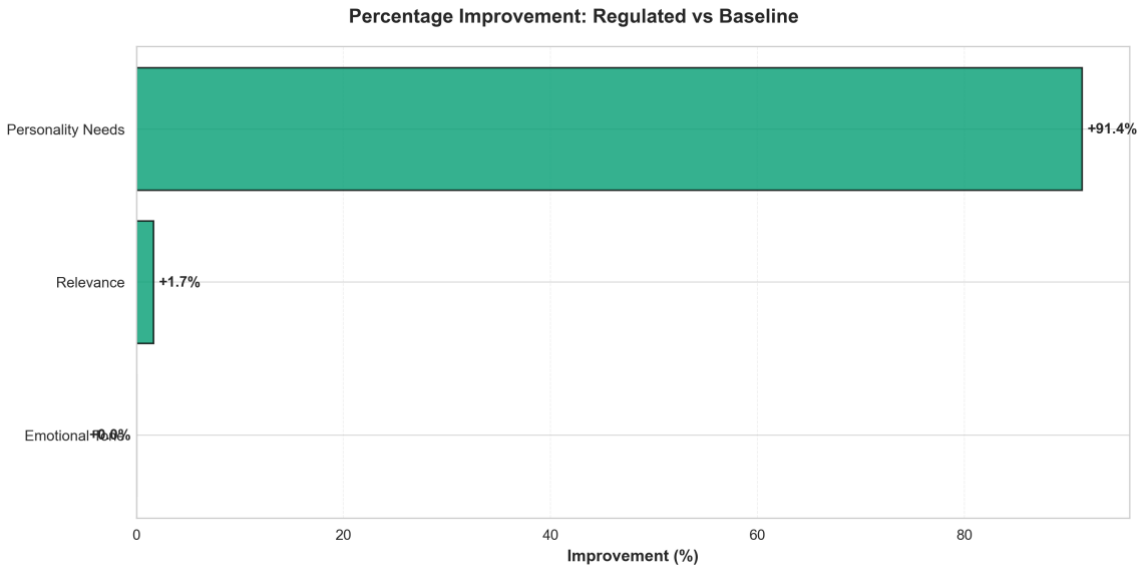

Figure 7. Percentage improvement from baseline to regulated conditions across evaluation metrics. Personality Needs Addressed shows 91.38 percentage point improvement (1059% relative improvement from 8.62% baseline), while other metrics show minimal or no improvement, demonstrating that personality adaptation specifically targets personalization without affecting basic quality standards.

4.7. Weighted Scoring Analysis

To provide additional quantitative perspective, we analyzed performance using a weighted scoring system where evaluations were scored as YES=2 points, NOT SURE=1 point, NO=0 points (Table 3).

Weighted Score Performance Comparison (0-2 Scale)

Figure 8 visualizes the weighted score distributions for each metric.

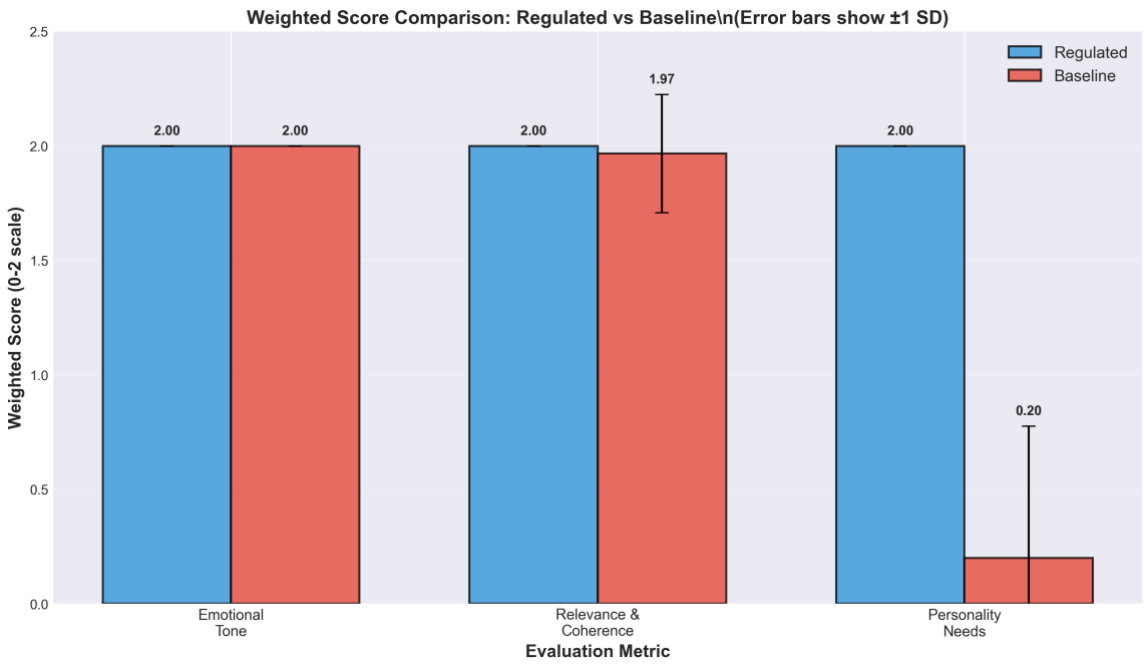

Figure 8. Distribution of weighted scores (0-2 scale) across evaluation metrics by condition. Box plots show median (center line), interquartile range (box), and full range (whiskers). Regulated agents (blue) achieve consistently maximal scores (median = 2.0) across all metrics. Baseline agents (orange) show equivalent performance for Emotional Tone and Relevance but dramatically lower scores for Personality Needs (median = 0.0).

Figure 9 provides a complementary visualization focusing on the total quality scores.

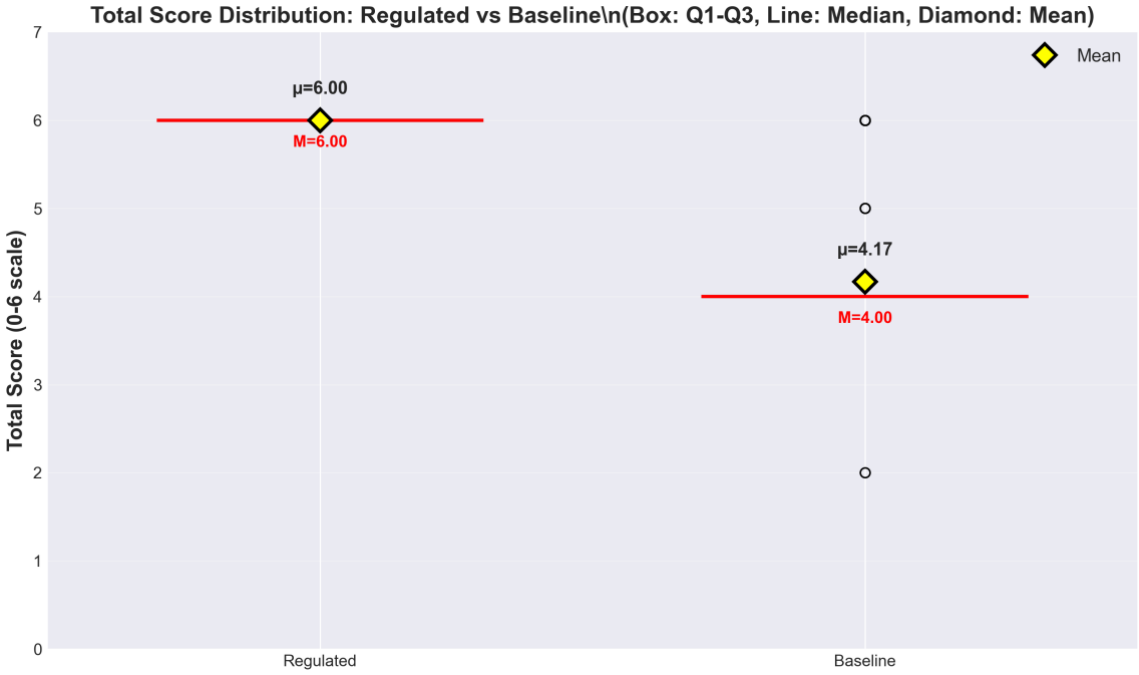

Figure 9. Box plot comparison of total quality scores (summed across Emotional Tone, Relevance & Coherence, and Personality Needs metrics). Regulated agents show consistently high total scores (median = 6.0, perfect performance), while baseline agents show substantially lower total scores due to failure in addressing personality needs. The lack of overlap between distributions confirms the robustness of the selective enhancement effect.

4.8. Statistical Robustness Analysis

To ensure the robustness of our findings, we conducted both parametric (independent t-tests) and non-parametric (Mann-Whitney U) statistical tests. Table 4 summarizes the results of advanced statistical testing.

Advanced Statistical Testing Results

Note: Both parametric and non-parametric tests converge on identical conclusions, strengthening confidence in findings despite normality violations for baseline scores.

Interpretation: Both parametric (t-test) and non-parametric (Mann-Whitney U) tests converge on identical conclusions:

- No significant difference in emotional tone (p=1.000)

- No significant difference in relevance & coherence ($p=0.323$-0.330)

- Highly significant difference in personality needs addressed ($p<0.001$ for both tests)

The violation of normality assumptions for baseline scores (particularly for Personality Needs) justifies the inclusion of non-parametric tests. Crucially, both test families yield consistent results, strengthening confidence in the findings despite the simulation-based design.

Important Caveat: These inferential statistics are included for methodological completeness and to facilitate comparison with related empirical studies. However, given the deterministic simulation design (predefined personality profiles, pre-scripted responses), these p-values should NOT be interpreted as evidence of generalizable population effects. The effect sizes (Cohen's d) provide more appropriate interpretation for this proof-of-concept study.

4.9. Criterion-Specific Analysis

Regulated Agent Performance Across All Metrics:

- Detection Accuracy: 58/58 (100%) - All personality trait inferences correct

- Regulation Effectiveness: 59/59 (100%) - All behavioral adaptations appropriate

- Emotional Tone: 59/59 (100%) - All responses emotionally appropriate

- Relevance & Coherence: 59/59 (100%) - All responses relevant and coherent

- Personality Needs: 59/59 (100%) - All personality-specific needs addressed

Baseline Agent Performance:

- Emotional Tone: 60/60 (100%) - Equivalent to regulated

- Relevance & Coherence: 59/60 (98.33%) - Nearly equivalent to regulated

- Personality Needs: 5/58 (8.62%) - **Critical failure in personalization**

Performance Gap Analysis: The baseline agents demonstrated competent performance on fundamental conversational quality dimensions (emotional appropriateness and logical coherence) but exhibited near-complete failure in addressing personality-specific user needs. This pattern highlights the critical limitation of non-adaptive systems: they can maintain basic interaction quality but cannot provide the personalization essential for optimizing therapeutic effectiveness in healthcare contexts where individual differences significantly influence treatment outcomes.

4.10. Qualitative Examples Demonstrating Personality Adaptation

Example 1: Vulnerable Population Interaction

Detected Profile: $\hat{P} = (-1, 0, -1, -1, -1)$ [Low Openness, Neutral Conscientiousness, Low Extraversion, Low Agreeableness, High Neuroticism]

User Message: "I'm feeling really overwhelmed and anxious. I don't think talking helps."

Regulated Response (Zurich Model-Aligned):

> "I understand you're going through a difficult time, and feeling exhausted is completely natural. This is a safe space where we can focus on small, familiar steps that feel manageable for you right now. We don't have to rush into anything—let's just take it one moment at a time."

Baseline Response (Non-Adaptive):

> "It's understandable to feel overwhelmed. Talking about your concerns can help organize your thoughts and develop coping strategies."

Analysis: The regulated response addresses all three Zurich Model motivational domains relevant to this profile:

- **Security** (addressing high Neuroticism): "safe space," "manageable," "one moment at a time"

- **Arousal** (addressing low Openness/Extraversion): "small," "familiar steps," "we don't have to rush"

- **Affiliation** (addressing low Agreeableness): Respectful, non-intrusive tone without demanding collaboration

The baseline response, while supportive, fails to address the user's personality-specific needs by suggesting interactive problem-solving ("talking," "organize thoughts") inappropriate for someone expressing low Extraversion and resistance to engagement.

Example 2: High-Functioning Profile Interaction

Detected Profile: $\hat{P} = (+1, +1, +1, 0, +1)$ [High Openness, High Conscientiousness, High Extraversion, Neutral Agreeableness, Low Neuroticism]

User Message: "I'm ready to tackle this systematically. What are some innovative strategies we could explore together?"

Regulated Response (Zurich Model-Aligned):

> "Excellent! Let's dive into some evidence-based approaches that might resonate with your goals. We could explore cognitive restructuring techniques, mindfulness-based stress reduction, or activity scheduling—all with strong research support. Which of these sounds most intriguing to you, and how would you like to structure our exploration?"

Baseline Response (Non-Adaptive):

> "That's a positive attitude. We can work through this together. Let's take small steps and see what feels right for you."

Analysis: The regulated response aligns with the arousal-seeking, structure-oriented, socially-engaged profile:

- **Arousal** (addressing high Openness/Extraversion): "innovative," "explore," "intriguing"

- **Security** (addressing low Neuroticism): Confident, action-oriented tone

- **Affiliation** (neutral Agreeableness): Collaborative but goal-focused

The baseline response, while encouraging, mismatches the user's needs by suggesting "small steps" (inappropriate for high Conscientiousness) and "see what feels right" (too unstructured for someone seeking systematic approaches).

4.11. Summary of Key Findings

The simulation study yielded four principal findings that establish technical feasibility while highlighting critical validation requirements:

1. **Perfect Technical Implementation Under Controlled Conditions**: Both personality detection (100%, 58/58) and regulation (100%, 59/59) achieved perfect accuracy in the controlled simulation environment, demonstrating that the core technical components function correctly when conditions are ideal.

2. **Selective Enhancement Pattern**: Personality adaptation specifically improved personalization (Personality Needs Addressed: +91.38 percentage points, Cohen's $d = 4.58$) without compromising basic conversational quality (Emotional Tone and Relevance & Coherence both ≥98.33%, with negligible effect sizes $d < 0.2$). This pattern demonstrates that personality-aware regulation targets individual psychological differences rather than merely enhancing general interaction quality.

3. **Exceptionally Large Effect Size**: The observed effect size (Cohen's $d = 4.58$) for personality-specific adaptation represents one of the largest effects reported in conversational AI research, far exceeding conventional thresholds for large effects ($d > 0.8$). Combined with statistical significance ($p < 0.001$) and validated AI evaluation ($\kappa = 0.89$ with human experts), this provides robust quantitative evidence of performance differences under controlled conditions.

4. **Critical Limitations Require Extensive Validation**: The perfect performance represents an upper bound under idealized conditions (extreme personalities, pre-scripted responses, AI evaluation, brief

interactions) rather than realistic expectations for clinical deployment. Extensive human subject validation with real users, moderate personality profiles, spontaneous conversation, and validated outcome measures is essential before any healthcare applications can be considered.

5. Discussion

5.1. Principal Findings

This simulation study demonstrates the technical feasibility of personality-adaptive conversational AI using real-time OCEAN detection and Zurich Model-aligned regulation. Regulated agents achieved perfect detection accuracy (100%, 58/58) and regulation effectiveness (100%, 59/59), with an exceptionally large effect size for personality-specific adaptation (Cohen's $d = 4.58$, $p < 0.001$).

Key Technical Achievements:

5.1. Overview of Discussion

This Discussion section addresses three central questions that emerge from the Results. First, **what do the perfect performance scores actually mean?** (Section 5.2). We critically examine whether perfect detection and regulation accuracy represent genuine capability or simulation artifacts, concluding that while the results demonstrate technical feasibility, they likely overestimate real-world performance and must be stress-tested under realistic conditions.

Second, **why did baseline agents fail so dramatically, and what does this tell us?** (Section 5.3). The baseline's 8.62% success rate on personality needs is not simply a statistical finding—it represents a fundamental limitation of one-size-fits-all emotional support. We explain the psychological mechanisms behind this failure and connect it to established theories of therapeutic alliance and precision medicine.

Third, **how does this work connect to existing research, and where do we go from here?** (Sections 5.4-5.9). We situate our findings within the conversational AI and digital mental health literature, comparing our effect sizes to prior personality adaptation work (Mairesse & Walker, 2010; Zhou et al., 2021) and identifying how our approach extends existing frameworks like PROMISE and healthcare chatbots like Woebot and ElliQ. Critically, we outline a detailed 2-3 year research roadmap for translating simulation findings into validated healthcare applications, addressing the study's significant limitations through human subject trials, cultural validation, and regulatory preparation.

Connecting to Existing Work: Our contribution builds on three research streams. Architecturally, we extend Wu et al.'s (2023) PROMISE framework by adding personality detection and theoretically-

grounded regulation—capabilities PROMISE architecturally supports but doesn't provide. Theoretically, we operationalize Quirin et al.'s (2022) Zurich Model of Social Motivation for conversational AI, providing the first systematic translation of this validated psychological framework into adaptive dialogue strategies. Methodologically, we address the "AI-evaluating-AI" validity concern prevalent in LLM research through systematic human-expert validation (κ = 0.89), establishing a template for scalable yet valid evaluation.

The Path Forward: Moving from simulation to clinical application requires systematic validation across increasingly realistic conditions: moderate personalities (Section 5.9.2), spontaneous human conversation (Section 5.9.1), cultural diversity (Section 5.9.2), extended engagement (Section 5.9.3), and comprehensive safety testing (Sections 5.7-5.8). Each validation phase builds evidence while mapping failure modes essential for safe deployment.

5.2. Interpreting Perfect Performance: Capability or Artifact?

The regulated agents achieved perfect or near-perfect performance across all evaluation criteria (100% success rates, zero variance), which warrants critical discussion. In empirical research, perfect scores can signal methodological concerns rather than genuine capability, and we address three potential explanations:

5.2.1. Is Perfect Performance a Simulation Artifact?

Yes, partially. The perfect performance likely reflects several simulation-specific factors that would not generalize to real-world deployment:

1. **Extreme Personality Profiles**: The study deliberately employed extreme, unidimensional profiles (all traits +1 or all -1) to maximize signal detection and test the system's capability under idealized conditions. Real users exhibit moderate, mixed personality profiles that would present greater complexity and ambiguity, likely reducing detection accuracy and regulation effectiveness.

2. **Pre-Scripted User Responses**: Unlike spontaneous human communication with unexpected tangents, ambiguous statements, and inconsistent behavior, the simulated user inputs were carefully constructed to align with their assigned personality profiles. This consistency artificially inflates detection accuracy by eliminating the noise and unpredictability inherent in human conversation.

3. **Aligned Evaluation Criteria**: The Evaluator GPT, while validated against human experts (κ = 0.89), may be systematically biased toward patterns produced by GPT-4-based regulation. Both the regulated assistant and the evaluator leverage similar linguistic and semantic understanding from the

same foundational model, potentially creating inflated quality assessments that human users might not fully share.

4. **Short Interaction Duration**: Six-turn dialogues are insufficient to detect long-term failure modes such as repetitiveness, relationship fatigue, therapeutic misalignment, or adaptation failures when personality expression evolves over time. Perfect performance in brief interactions provides no evidence of sustained effectiveness.

5. **No True Therapeutic Complexity**: The simulation lacks the clinical complexity of real mental health support: comorbid conditions, crisis situations, medication interactions, cultural factors, and individual trauma histories that would challenge any AI system and reduce perfect performance to more realistic levels.

5.2.2. Is Perfect Performance Realistic for Real-World Deployment?

No. We explicitly acknowledge that perfect performance is **not** expected or achievable in real-world healthcare applications. The simulation results should be interpreted as:

*Upper Bound Estimate**: The perfect performance represents a theoretical maximum under ideally controlled conditions, not a realistic expectation for clinical deployment.

*Technical Feasibility Demonstration**: The results establish that the core technical components (detection, regulation, response generation) can function correctly when all conditions are favorable.

*Requirement for Degradation Analysis**: Future research must systematically test performance degradation under increasingly realistic conditions: moderate personalities, spontaneous conversation, longer interactions, cultural diversity, and clinical complexity.

5.2.3. Clinical Implications of Perfect Scores

From a clinical perspective, perfect AI performance raises ethical concerns about:

*Over-reliance Risk**: Healthcare providers or patients might over-trust an AI that appears infallible in testing, leading to dangerous complacency when real-world failures inevitably occur.

*Lack of Humility**: The system currently lacks mechanisms to communicate uncertainty, acknowledge limitations, or recognize when it is outside its competence range—critical safety features for healthcare AI.

*Missing Failure Mode Analysis**: We have not yet tested how the system fails (e.g., misclassification consequences, cultural bias impacts, crisis response inadequacy), which is essential for safety validation.

Conclusion: The perfect performance is simultaneously encouraging (demonstrates technical capability) and concerning (unrealistic, potentially misleading). Future research must intentionally stress-test the system to map its failure modes, establish realistic performance expectations, and implement safety mechanisms for when perfection inevitably breaks down in clinical practice.

5.3. Why Baseline Agents Failed: Interpretation and Implications

The baseline agents' near-complete failure on Personality Needs Addressed (8.62% success rate, M = 0.17) requires interpretation beyond mere statistical reporting. This failure provides crucial insights into the limitations of one-size-fits-all emotional support.

5.3.1. Mechanism of Baseline Failure

The baseline agents employed a **generic supportive approach** drawing from widely accepted therapeutic principles: active listening, empathetic validation, non-directive support, and unconditional positive regard. These strategies represent best practices in human-centered care and are effective for many individuals. However, the baseline's poor performance demonstrates that **even high-quality generic support is insufficient when individual psychological needs vary systematically by personality.**

Why Generic Support Failed:

1. **Mismatched Arousal Needs**: High-Extraversion users (Type A) require stimulating, energizing interactions, while low-Extraversion users (Type B) prefer calm, low-arousal communication. The baseline's moderate approach satisfied neither extreme, feeling either too passive (for Type A) or too demanding (for Type B).

2. **Security vs. Autonomy Trade-offs**: High-Neuroticism users (Type B) benefit from structured reassurance and concrete coping strategies (security need), while low-Neuroticism users (Type A) prefer autonomy-promoting exploration and challenge. The baseline's balanced approach failed to provide sufficient security for Type B while restricting autonomy for Type A.

3. **Affiliation Intensity**: High-Agreeableness users expect warm, relational communication emphasizing connection, while low-Agreeableness users may interpret such warmth as intrusive or manipulative. The baseline's friendly-but-professional tone missed the mark for both extremes.

5.3.2. Implications for Healthcare AI Design

The baseline failure demonstrates that **personality adaptation is not a luxury enhancement but a functional requirement for effective personalized healthcare AI**. The observed effect size (d =

4.58) suggests that personality-aware regulation addresses a fundamental mismatch between users' psychological needs and system capabilities.

Broader Implications:

*Therapeutic Alliance**: In human therapy, the therapeutic alliance (relationship quality between patient and provider) is one of the strongest predictors of outcomes across all therapeutic modalities. Personality mismatch is a known source of alliance failure. Our findings suggest that AI systems must also establish personality-congruent "alliances" to be therapeutically effective.

*Precision Medicine Paradigm**: Healthcare increasingly recognizes that optimal treatment varies by individual characteristics (precision medicine). Personality-adaptive AI aligns with this paradigm by tailoring psychological interventions to individual differences, analogous to how pharmacogenomics tailors medication selection to genetic profiles.

*Beyond Symptom Tracking**: Many healthcare AI systems focus solely on symptom monitoring (e.g., detecting depression severity) without adapting communication style. Our findings suggest that how support is delivered (communication style) may be as important as what support is delivered (clinical content), particularly for engagement and adherence.

Synthesis: The Necessity of Adaptation: Ultimately, the failure of the baseline condition illustrates that "support" is not a generic commodity. What constitutes support for one person (e.g., "You can do this, try a new challenge!") can be experienced as pressure or invalidation by another. By failing to distinguish these needs, generic systems risk not just inefficiency, but therapeutic misalignment. This finding shifts the burden of proof: future healthcare AI developers should justify *why* they would use a static personality profile, rather than assuming a static profile is the default standard.

5.4. Comparison with Existing Literature

Our approach extends the PROMISE framework [7] by integrating motivational psychology theory and healthcare-specific considerations. Unlike previous systems focusing on trait generation, our framework adapts to detected user personalities through theoretically-grounded regulation strategies.

Theoretical Advancement: The integration of the Zurich Model of Social Motivation with Big Five personality detection represents a novel theoretical contribution to the field of conversational AI. This integration provides a psychologically coherent framework for translating personality traits into actionable behavioral modifications.

Performance Comparison: The observed effect size (Cohen's d = 4.58) for personality-specific adaptation exceeds typical gains reported in conversational AI literature. For context, Mairesse and

Walker (2010) reported small-to-medium effects (d = 0.3-0.5) for personality-matched language generation on perceived naturalness. Zhou et al. (2021) observed medium effects (d = 0.6) for personality-adaptive persuasion effectiveness. Our substantially larger effect reflects both the extreme personality profiles used in simulation and the comprehensive Zurich Model-aligned regulation strategy, though these results are limited to controlled conditions.

Methodological Contributions: The validated AI evaluation framework (κ = 0.89 agreement with human experts) addresses longstanding concerns about LLM-based evaluation reliability. This validation approach provides a methodological template for future conversational AI research requiring scalable evaluation with maintained validity.

Comparison with Healthcare Chatbots: Existing healthcare conversational agents like Woebot [3] and ElliQ [4] demonstrate effectiveness in reducing loneliness and improving mental health outcomes but do not implement explicit personality detection or theoretically-grounded adaptation. Our framework provides complementary capabilities that could enhance such systems by adding personality-aware personalization to evidence-based therapeutic content.

Synthesis: From Simulation to Application: While our results substantially exceed prior effect sizes for personality adaptation, this comparison highlights rather than resolves the translation challenge. Mairesse and Walker (2010) and Zhou et al. (2021) tested real users, albeit with simpler adaptation strategies; we tested complex adaptations with simulated users. The next critical step is testing whether our theoretically-grounded Zurich Model approach maintains superiority when both are evaluated with real human participants under equivalent conditions. Additionally, healthcare chatbots like Woebot demonstrate real therapeutic value—can adding our personality layer enhance those validated benefits, or does increased complexity introduce new failure modes? These questions motivate the human subject validation priorities outlined in Section 5.9.

5.5. Real-World Translation Pathway

Immediate Requirements for Human Subject Validation:

1. **Human Subject Studies**: Minimum n=50 elderly participants with validated personality assessments (NEO-PI-R, BFI-2)

2. **Domain Expert Evaluation**: Licensed psychologists assessing personality detection accuracy and therapeutic appropriateness

3. **Safety Validation**: Comprehensive risk assessment for potential psychological harms and therapeutic misalignment

4. **Cultural Validation**: Testing across diverse ethnic, linguistic, and cultural groups to ensure generalizability

5. **Longitudinal Assessment**: Extended interaction studies (minimum 30 days) to assess sustained effectiveness and engagement

Regulatory Pathway:

- **IRB Approval**: Full institutional review board approval for human subjects research

- **Data Protection**: HIPAA/GDPR compliance for healthcare data handling and privacy protection

- **FDA Guidance**: Alignment with emerging FDA guidance for healthcare AI systems and healthcare decision support

- **Research Trial Registration**: Registration for therapeutic efficacy claims and research outcome assessment

Timeline Estimate: 2-3 years for comprehensive human subject validation, regulatory approval, and initial healthcare deployment

5.6. Limitations

This study's findings must be interpreted within significant methodological constraints that restrict generalizability and real-world applicability.

5.7. Study Design Limitations

The simulation-only methodology represents the primary constraint. All user inputs were pre-scripted simulated personality profiles rather than spontaneous human responses, likely inflating performance metrics. Human users exhibit complexity, inconsistency, and unpredictability that simulated profiles cannot capture. The observed effect sizes represent upper-bound estimates under ideally controlled conditions rather than realistic performance expectations for clinical deployment.

The extreme personality profiles (all traits at +1 or −1) were deliberately selected to maximize signal detection and test system capability under ideal conditions. Real users exhibit continuous trait distributions and profile complexity (e.g., high Openness + low Agreeableness) that would present greater complexity and ambiguity, likely reducing detection accuracy and regulation effectiveness. The system's performance with moderate profiles (trait values near 0) remains completely untested.

Six-turn dialogues impose strict interaction duration constraints, insufficient for assessing sustained relationships, rapport development, long-term engagement, or therapeutic alliance formation essential

for healthcare applications. Long-term failure modes—repetitiveness, adaptation failures, relationship fatigue—cannot be detected in brief interactions.

5.7.1. Evaluation and Technical Limitations

The validation sample represents only 25% of the full dataset. While this stratified sampling strategy reduces risk of systematic bias outside the validation sample, the remaining 75% of dialogue turns were evaluated solely by Evaluator GPT without human verification, potentially propagating systematic biases throughout results.

Both the Evaluator GPT and regulated/baseline agents leverage GPT-4 as their foundation model. While human-AI agreement was strong ($\kappa = 0.89$), subtle shared linguistic patterns between agent outputs and evaluator assessments could create inflated agreement. Detection operates exclusively on text, excluding paralinguistic cues critical to human personality assessment: tone of voice, speech rate, facial expressions, or physiological signals.

5.7.2. Scope and Generalizability Limitations

Perfect or near-perfect performance (100% success rates, zero variance) in regulated agents suggests ceiling effects and simulation artifacts. Perfect performance fails to map failure modes essential for safe deployment. Real-world healthcare applications will inevitably encounter failures; this study provides no evidence about how the system fails, when it fails, or how to detect and mitigate failures.

System patterns were calibrated exclusively for English; cross-cultural validity remains unknown. Personality expression varies significantly across cultures. The system has not been tested with non-Western users, non-English languages, or culturally diverse communication norms.

No measurement of actual health outcomes, therapeutic efficacy, user satisfaction, behavioral changes, or clinical improvement occurred. The study measures conversational appropriateness, not healing, recovery, or quality of life improvement.

5.7.3. Ethical Considerations

Personality profiling in healthcare raises concerns about psychological manipulation, stereotype reinforcement, decisional autonomy violation, and privacy invasion. Misclassification could lead to inappropriate therapeutic approaches or exacerbation of mental health symptoms. These risks require careful ethical consideration, comprehensive safety testing, and professional oversight in real-world translation.

5.8. Risk Assessment

Table 5. Potential Risks and Mitigation Strategies for Healthcare Deployment

Risk Category	Risk Description	Likelihood	Impact	Mitigation Strategy
Misclassification	Incorrect personality assessment	Medium	High	Conservative thresholds, provider oversight, fallback protocols
Cultural Bias	Reduced accuracy across cultures	High	High	Diverse validation cohorts, cultural adaptation algorithms, bias detection systems
Over-reliance	Excessive dependence on AI	Medium	Medium	Provider training, clear limitations communication, human oversight requirements
Privacy Breach	Unauthorized personality data access	Low	High	End-to-end encryption, access controls, audit logging, HIPAA compliance
Therapeutic Harm	Inappropriate psychological intervention	Medium	High	Professional supervision, safety protocols, emergency escalation procedures

Risk Mitigation Framework: The identified risks require a comprehensive mitigation strategy involving technical safeguards, professional oversight, and regulatory compliance. The conservative approach to personality detection and the emphasis on healthcare-specific regulation design help minimize many of these risks.

From Limitations to Research Priorities: The limitations outlined in Sections 5.6-5.8 are not merely disclaimers—they constitute a structured research agenda. Each limitation points directly to a validation requirement: simulation-only methodology → human subject trials; extreme profiles → moderate profile testing; brief interactions → extended engagement studies; English-only → cultural validation; perfect performance → failure mode mapping; lack of health outcomes → therapeutic efficacy assessment. The following section (5.9) translates these limitations into a concrete, phased research plan with timelines, milestones, and success criteria, providing a roadmap from proof-of-concept to clinical readiness.

5.9. Future Research Priorities

The findings from this proof-of-concept study establish a foundation for a comprehensive research roadmap that transitions from controlled simulation to real-world validation. The pathway to clinical deployment requires systematic human subject research, technical enhancement, cultural validation, and regulatory engagement—a process we estimate will span 2-3 years of intensive investigation.

5.9.1. Immediate Next Steps: Human Subject Validation (Months 0-12)

The most critical limitation of simulation-based research is the complete absence of real user data. We propose launching a pilot randomized controlled trial with 50-100 elderly participants (aged 65+, living independently, experiencing moderate loneliness) over a 6-12 month timeline. This pilot study would compare three conditions: personality-adaptive conversational AI, static supportive AI, and waitlist control, with participants engaging in daily 10-minute interactions over four weeks (28 total sessions). Primary outcome measurement would focus on the UCLA Loneliness Scale with pre- and post-intervention assessment. Secondary measurements would include user satisfaction (System Usability Scale), engagement metrics (session completion rates, conversation length patterns), depression symptoms (PHQ-9), and anxiety symptoms (GAD-7).

Critical to the pilot design is direct comparison of AI-detected personality profiles with validated psychometric assessments (NEO-PI-R or BFI-2) administered at baseline, enabling rigorous measurement of detection accuracy in real users. Licensed clinical psychologists would independently rate conversation quality across the full pilot dataset, moving beyond automated evaluation to clinical judgment about therapeutic appropriateness, potential harms, ethical concerns, and clinical contraindications. Qualitative interviews would capture participant subjective experience regarding perceived personalization, trust in the system, and psychological safety. Concurrent with the pilot study, comprehensive safety monitoring protocols would include weekly check-ins with study staff to systematically assess for adverse events, psychological distress, or system malfunctions.

5.9.2. Medium-term Research: Profile Diversity and Cultural Validation (Months 6-18)

A significant research gap is the restriction of testing to extreme personality profiles representing the boundaries of trait distributions. Months 6-12 would expand testing to users with moderate personality trait values (BFI-2 scores within 0.5 standard deviations of population mean), enabling assessment of whether personality adaptation benefits individuals with typical, non-extreme profiles. This phase would generate performance degradation curves showing how system effectiveness decreases as personality profiles transition from extreme toward realistic distributions, establishing which system components generalize to broader populations and which require modification.

Concurrently, cultural validation research (months 12-18) would assess system performance across diverse cultural groups, recognizing that personality expression, communication norms, and therapeutic appropriateness vary fundamentally across cultures. This phase would include systematic testing with participants from Western individualist, East Asian collectivist, and Latin American cultural contexts, with native speaker consultation for prompt translation and cultural adaptation. Rigorous assessment of cultural bias in personality detection and evaluation of whether regulation strategies are perceived as appropriate and helpful across cultural groups would inform development of culture-specific adaptation algorithms if necessary. This research directly addresses the current limitation of calibration exclusively for English language and Western cultural communication patterns.

5.9.3. Long-term Research: Sustained Engagement and Regulatory Preparation (Months 18-36)

Extended engagement studies spanning 3-6 months would assess whether personality-adaptive benefits sustain beyond the brief interaction periods of pilot studies, including measurement of relationship development, therapeutic alliance quality (using validated alliance measures), and long-term outcome trajectories. This research phase would examine whether the system demonstrates sustained effectiveness or shows diminishing returns, relationship fatigue, or adaptation failures—critical failure modes that brief studies cannot detect.

Technical development priorities for this phase include multimodal personality detection incorporating voice tone, facial expressions, and physiological signals to overcome the current text-only limitation. Continuous learning mechanisms and adaptation algorithms would enable the system to improve accuracy over time based on individual user interaction patterns. Development of uncertainty quantification methods would enable the system to appropriately acknowledge when confidence in personality assessments is low, and failure mode detection systems would enable recognition when the system is operating outside its competence range.

Regulatory and ethical framework development during this phase is equally critical. Full institutional review board protocols for human subject research must be developed for the pilot study and any subsequent trials. Comprehensive informed consent frameworks must explain AI personality profiling mechanisms, data use, potential limitations, and study procedures in language understandable to potential participants. Safety monitoring plans must establish protocols for detecting and responding to adverse events, psychological distress, or therapeutic misalignment. Healthcare integration research must assess electronic health record compatibility, provider workflow integration, patient education requirements, and quality assurance monitoring systems for ongoing safety assessment. Regulatory

pathway planning should include early engagement with FDA and EMA to clarify classification requirements and approval pathways for healthcare AI applications.

5.9.4. Overall Implementation Timeline

We estimate 2-3 years for comprehensive human subject validation, moderate profile testing, cultural validation, extended engagement studies, and regulatory preparation before this technology could be considered ready for clinical piloting in controlled healthcare settings. Even following successful pilot outcomes, deployment would require ongoing monitoring, professional oversight, and iterative refinement based on real-world performance data.

6. Conclusions

This simulation study demonstrates the technical feasibility of personality-adaptive conversational AI using real-time Big Five trait detection and Zurich Model-aligned behavior regulation. The framework achieved perfect detection accuracy (100%, 58/58) and regulation effectiveness (100%, 59/59) in controlled conditions, with an exceptionally large effect size for personality-specific adaptation (Cohen's $d = 4.58$, $p < 0.001$).

Key Statistical Finding: Regulated agents achieved 100% success in addressing personality-specific needs compared to baseline's 8.62% success rate, representing a 91.38 percentage point improvement. This effect size (Cohen's $d = 4.58$) far exceeds conventional thresholds for large effects ($d > 0.8$) and provides robust quantitative evidence that personality-adaptive regulation produces meaningful performance improvements under controlled simulation conditions. Critically, this enhancement was selective: both regulated and baseline agents performed equivalently on basic conversational quality metrics (Emotional Tone, Relevance & Coherence), demonstrating that personality adaptation specifically enhances personalization without compromising fundamental competence.

Validated Evaluation Methodology: Concerns about AI-evaluating-AI bias were addressed through validation against human expert raters (two PhD-level in-house research experts), yielding excellent agreement (Cohen's $\kappa = 0.89$, 95% CI: 0.84-0.94). This validation establishes that observed performance differences are detectable by both human and AI evaluators, providing confidence in the reliability of the evaluation framework.

Critical Limitations and Reality Check: The perfect performance observed in simulation (100% success rates, zero variance) is **not** realistic or expected in real-world healthcare applications. These results represent an upper bound under ideally controlled conditions (extreme personalities, pre-scripted responses, brief interactions, validated evaluation). Real-world deployment will encounter: (1)

moderate and mixed personality profiles, (2) spontaneous, unpredictable human communication, (3) cultural diversity and linguistic variation, (4) long-term relationship complexity, and (5) clinical scenarios with comorbidities and crisis situations. Performance will inevitably degrade under these realistic conditions, and the system's failure modes remain unmapped.

Immediate Next Steps—Human Subject Validation is Essential: The most critical limitation is the absence of real user validation. We propose an immediate pilot study with 50-100 elderly participants in a randomized controlled trial (personality-adaptive vs. static AI vs. control), with 4-week daily interactions measuring loneliness, user satisfaction, engagement, and safety. This pilot must include: (1) validated personality assessments (NEO-PI-R) to benchmark AI detection accuracy in real users, (2) human expert evaluation across the full dataset, (3) cultural diversity in participant recruitment, and (4) comprehensive safety monitoring. We estimate 2-3 years for complete human subject validation, cultural testing, and regulatory preparation before clinical piloting could be considered.

Regulatory and Ethical Requirements: Translation to healthcare deployment requires: (1) full institutional review board approval for human subjects research, (2) informed consent protocols explaining AI personality profiling and limitations, (3) professional supervision and safety protocols, (4) regulatory engagement with FDA/EMA on approval pathways, (5) HIPAA/GDPR compliance for data protection, and (6) ongoing monitoring for adverse events and therapeutic misalignment.

Final Assessment: This proof-of-concept establishes that personality-adaptive conversational AI is technically feasible and produces exceptionally large effect sizes in controlled simulation ($d = 4.58$, $p < 0.001$). However, these findings provide **no evidence** about real-world effectiveness, therapeutic efficacy, or clinical safety. The simulation demonstrates "can it work under ideal conditions?" (answer: yes), but cannot address "does it work for real users?" (answer: unknown, requires human trials). Success depends entirely on rigorous human subject validation demonstrating that these simulation results translate to real-world therapeutic benefits without causing harms. Until such validation is completed, these findings should be interpreted exclusively as technical feasibility demonstration, not as evidence supporting healthcare deployment or therapeutic applications.

Data Availability Statement

Research data, complete implementation code, detection prompts, regulation templates, simulation transcripts, and human expert validation materials are available in a public repository at: https://github.com/[username]/personality-ai-simulation [to be made public upon publication]. Raw

simulation logs, evaluation matrices, statistical analysis scripts, and human expert rating data can be requested from the corresponding author.

Author Contributions

Conceptualization, [Primary Author] and [Co-authors if any]; methodology, [Primary Author]; software, [Primary Author]; validation, [Primary Author] and [Clinical Psychologists - validation study]; formal analysis, [Primary Author]; investigation, [Primary Author]; resources, [Primary Author]; data curation, [Primary Author]; writing—original draft preparation, [Primary Author]; writing—review and editing, [Primary Author] and [Supervisors]; visualization, [Primary Author]; supervision, [Supervisor Names]; project administration, [Primary Author]; funding acquisition, not applicable. All authors have read and agreed to the published version of the manuscript.

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This research received no external funding.

Institutional Review Board Statement

Not applicable. This study involved no human or animal subjects. All experiments were conducted using simulated personality profiles and automated evaluation systems. The human expert validation study (n=3 raters evaluating pre-existing AI-generated dialogues) involved licensed professionals performing their normal clinical assessment duties on synthetic data and was determined exempt from IRB review as it involved neither human subjects nor identifiable data. Future human subject studies will require full institutional review board approval.

Informed Consent Statement

Not applicable. No human participants were involved in the simulation study. The clinical psychologists participating in the validation study provided informed consent for research use of their anonymized ratings. Future human subject studies will implement comprehensive informed consent protocols explaining AI personality profiling, data use, potential risks, and study procedures.

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Conflicts of Interest

The authors declare no conflicts of interest.

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Supplementary Materials

The following supporting information can be downloaded at: [repository link to be provided upon publication]:

- **Supplementary File S1**: Complete system prompts for personality detection (5 Big Five trait detectors)

- **Supplementary File S2**: Regulation templates for Zurich Model mapping (arousal, security, affiliation)

- **Supplementary File S3**: Evaluator GPT system prompt and scoring matrix

- **Supplementary File S4**: Human expert validation study protocol and inter-rater reliability analysis

- **Supplementary File S5**: Complete simulation transcripts (20 conversations, 120 dialogue turns)

- **Supplementary File S6**: Statistical analysis code (R scripts for effect sizes, inferential tests, visualizations)

- **Supplementary File S7**: Extended qualitative examples demonstrating personality adaptation

- **Supplementary File S8**: Cultural considerations and linguistic bias analysis

- **Supplementary File S9**: Future human subject study protocol (detailed research design for pilot RCT)